

Almost-idempotent positive maps and approximate Jordan algebras

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(collaborative working report)

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Abstract

Kitaev showed that a quantum channel which is almost idempotent — a unital completely positive map Φ on the operators of a finite-dimensional Hilbert space with $\|\Phi^2 - \Phi\|_{\text{cb}} \leq \eta$ — carries, on its near-fixed points, the structure of an approximate C^* -algebra, and that every finite-dimensional approximate C^* -algebra is close to a genuine one. We study the same question when Φ is only assumed *positive*, not completely positive. We show that the correct invariant is then a *Jordan* algebra rather than an associative one, for a precise structural reason: a positive unital map need not satisfy the Schwarz inequality $\Phi(x)^*\Phi(x) \leq \Phi(x^*x)$, but it always satisfies Kadison’s inequality $\Phi(x)^2 \leq \Phi(x^2)$ for self-adjoint x , and the latter involves exactly the data of a Jordan algebra.

Our main complete result is a *bridge theorem*: for any unital positive Φ with $\|\Phi^2 - \Phi\| \leq \eta$, the near-fixed-point space $A = \text{Im } P$ of the spectral idempotent $P = \theta(2\Phi - 1)$, equipped with the product $a \bullet b = P(a \circ b)$, satisfies the axioms of a Jordan algebra up to an error $O(\sqrt{\eta})$, with constants independent of dimension. The proof uses no dilation and no complete positivity; the key fact is that the “square holes” $P(a^2) - a^2$ are almost-positive null elements, so that quadratic error terms are $O(\eta)$ while the exact Jordan identity of the ambient algebra cancels the leading term.

We then carry the programme as far as it currently goes and report honestly on what is proved, conditional, or open. (i) We reduce *exact* factorization by unital positive maps to a single stability question — that a near-positive idempotent is $O(\sqrt{\eta})$ -close to a genuine positive one — proving the factorization conditional on it and showing that no generic positivity-rounding can replace it. (ii) In the commutative case this becomes a stability problem for almost-idempotent stochastic matrices; we record the sharp square-root exponent, several proved special cases, and an exact commutative factorization theorem, leaving one geometric lemma open. (iii) For the abstract structure theorem — every finite-dimensional approximate Jordan algebra is close to a genuine one — we present a dimension-free splitting of the Jordan cohomology coboundary for exact cocycles on every finite-dimensional JB-algebra (the exact-adjoint benchmark), isolating the approximate-cocycle and positivity gaps that remain. (iv) We analyse the exponent (1/2 in general; 1 for completely positive maps, and conjecturally for decomposable maps, conditional on a dilation-compatibility hypothesis that we make precise). The report is a consensus document of two collaborating authors; every statement is tagged as a definition, a cited fact, a proved result, a conditional result, or an open target.

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1 Introduction

The question. A physical system sometimes hides a smaller logical system inside it: the logical observables are encoded in the physical ones, and a recovery operation returns the system to the encoded subspace. Mathematically, such a recovery is a quantum channel T that is idempotent, $T^2 = T$. Kitaev [6] studied the realistic case where the recovery is only *approximately* idempotent, and asked what algebraic structure the encoded observables carry. He worked with the dual map Φ of the channel, a unital completely positive map on the algebra $B(\mathcal{H})$ of operators on a finite-dimensional Hilbert space, and measured non-idempotence by $\|\Phi^2 - \Phi\|_{\text{cb}} \leq \eta$. His answer: the near-fixed points of Φ form an *approximate C^* -algebra*, and every finite-dimensional approximate C^* -algebra is, with universal (dimension-free) constants, close to a genuine C^* -algebra.

This report. We ask the same question for maps that are only *positive*, not completely positive. This is the natural setting whenever one cares about distinguishability of states rather than full quantum operations, and it is the setting of van Luijk and Wilming [8], who show that for positive trace-preserving maps the relevant invariant is a *Jordan algebra*. The present report does the following, in order of completeness:

1. It records the consensus *definitions* (positive map, Jordan product, JB-algebra, order-unit space, approximate Jordan algebra). Standard background definitions are matched against local source copies; project definitions such as the ε -JB notion are marked (CONSENSUS) and tied to internal proof files in `PROVENANCE.md`.
2. It proves the *bridge theorem* (Section 6): from a unital positive almost-idempotent Φ one obtains, with no use of dilations or complete positivity, a space that satisfies the Jordan-algebra axioms up to $O(\sqrt{\eta})$. This is the positive-map analogue of the first half of Kitaev's construction, and it is complete. A companion result (Section 7) treats the ambient product under a faithful invariant state.
3. It reduces *exact factorization* by unital positive maps (Section 8) to one stability question about near-positive idempotents, proving the factorization conditional on it; in the *commutative* case (Section 9) this becomes a stability problem for almost-idempotent stochastic matrices, for which the sharp exponent, several proved special cases, and an exact factorization theorem are established.
4. It develops the *abstract structure theorem* (Section 10) — the analogue of the second half of Kitaev's theorem — proving a dimension-free splitting of the Jordan cohomology for exact cocycles on every finite-dimensional JB-algebra, and isolating the remaining gaps.
5. It analyses the *exponent* $\sqrt{\eta}$ versus η (Section 11) and the role of decomposability, and summarises the status throughout (Section 12).

Why Jordan, and not associative. The reason positivity forces a Jordan, rather than associative, structure can be stated in one line. Kitaev's verification of the C^* -norm axiom rests on the Schwarz inequality $\Phi(x)^*\Phi(x) \leq \Phi(x^*x)$, which holds for 2-positive (in particular completely positive) maps. A merely positive unital map need not satisfy it; what it does satisfy is Kadison's inequality $\Phi(x)^2 \leq \Phi(x^2)$ for self-adjoint x [5]. The latter involves only self-adjoint elements and squares — exactly the data on which a Jordan algebra is built. Replacing the associative product xy by the symmetric product $x \circ y = \frac{1}{2}(xy + yx)$ throughout is therefore not a convenience but a necessity, and the entire theory reorganises around it.

Conventions and status flags. All Hilbert spaces are finite-dimensional and complex. We write $B(\mathcal{H})$ for the operators on $\mathcal{H}$ and $B(\mathcal{H})_{\text{sa}}$ for the self-adjoint ones. Statements carry a status: (PROVED) for results with a complete proof in this report or transcribed from a verified source; (CITED) for results imported from the literature; (CONSENSUS) for definitions and conventions adopted by agreement between the authors; (CONDITIONAL) for implications whose stated hypothesis is still open; (BENCHMARK) for exact-cocycle or restricted-module results that do not yet prove the full target; (OBSTRUCTION) for proved no-go statements about a route; (SOURCE-EXTRACTED) for claims supported by local extraction but still awaiting primary byte-check; (NUMERICAL) for evidence that is not used as proof; and (OPEN) for targets of the programme not yet established. Constants denoted C are universal and never depend on $\dim \mathcal{H}$ unless explicitly stated.

Reading guide (and how to verify it). The fastest way to see what is established and what remains is the status ledger (Table 2) together with the dependency graph (Figure 1), both at the end of Section 12. One complete, unconditional result anchors the report — the bridge theorem (Theorem 6.1), Section 6 — and everything else is organised around it: an exact-factorization reduction (Sections 8 and 9), the abstract structure programme (Section 10), and the exponent question (Section 11). The three genuinely open problems are isolated in Section 12. Two further checkable layers sit under the prose. First, every statement imported from the literature carries a citation whose bibliography entry links to the source (arXiv, journal open access) where one is publicly available. Second, results marked with a green ‘✓ af-validated’ badge are machine-checked: the badge links to a self-contained formal workspace `proofs/<id>/` whose root statement is byte-identical to the result’s registry contract and whose every step was proved and independently verified. Thirteen results — the bridge and its lemmas, and the dilation-compatible theorem — are af-validated; they are the green nodes of Figure 1.

2 Preliminaries

This section fixes the analytic setting. Everything is finite-dimensional; the only norm is the operator norm, and the only positivity used is positivity of maps (never complete positivity).

Definition 2.1 (Operators, self-adjoint part, states). (CITED) Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space and $B(\mathcal{H})$ the algebra of linear operators on $\mathcal{H}$, with the operator norm $\|x\| = \sup_{\|\xi\| \leq 1} \|x\xi\|$. The *self-adjoint part* is $B(\mathcal{H})_{\text{sa}} = \{x \in B(\mathcal{H}) : x = x^*\}$, a real vector space. For $a \in B(\mathcal{H})_{\text{sa}}$ we write $a \geq 0$ if every eigenvalue of a is nonnegative; this induces the partial order $a \geq b \iff a - b \geq 0$, with unit $\mathbf{1}$ the identity operator. A *state* is a linear functional $\omega : B(\mathcal{H}) \rightarrow \mathbb{C}$ that is positive ($\omega(a) \geq 0$ for $a \geq 0$) and unital ($\omega(\mathbf{1}) = 1$).

Definition 2.2 (Positive, unital, and positive unital maps). (CITED) A linear map $\Phi : B(\mathcal{H}) \rightarrow B(\mathcal{H})$ is *positive* if $\Phi(a) \geq 0$ whenever $a \geq 0$; a positive map is automatically self-adjointness-preserving, $\Phi(a^*) = \Phi(a)^*$, so it restricts to a real-linear map $\Phi : B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$. It is *unital* if $\Phi(\mathbf{1}) = \mathbf{1}$. A *positive unital map* is a contraction on the self-adjoint part: for $a \in B(\mathcal{H})_{\text{sa}}$ one has $-\|a\| \mathbf{1} \leq a \leq \|a\| \mathbf{1}$, hence $-\|a\| \mathbf{1} \leq \Phi(a) \leq \|a\| \mathbf{1}$ by positivity and unitality, so $\|\Phi(a)\| \leq \|a\|$; with $\Phi(\mathbf{1}) = \mathbf{1}$ this gives $\|\Phi\| = 1$ on $B(\mathcal{H})_{\text{sa}}$.

The order structure of $B(\mathcal{H})_{\text{sa}}$ is what survives when the associative product is dropped; it is captured abstractly by the notion of an order-unit space.

Definition 2.3 (Order-unit space and order-unit norm). (CITED) A real vector space A with a proper convex cone A^+ (giving $a \geq b \iff a - b \in A^+$) and a distinguished element $e \in A^+$ is an

order-unit space if (i) e is an *order unit*: for each $a \in A$ there is $\lambda > 0$ with $-\lambda e \leq a \leq \lambda e$, and (ii) A is *Archimedean*: $na \leq e$ for all $n \in \mathbb{N}$ implies $a \leq 0$. The *order-unit norm* is

$$\|a\| = \inf\{t > 0 : -te \leq a \leq te\}.$$

With $e = \mathbf{1}$ and cone $A^+ = \{a \in B(\mathcal{H})_{\text{sa}} : a \geq 0\}$, the self-adjoint part $B(\mathcal{H})_{\text{sa}}$ is an order-unit space, and its order-unit norm coincides with the operator norm.

Remark 2.4. (PROVED) That $B(\mathcal{H})_{\text{sa}}$ is an order-unit space is immediate from Definition 2.3: for self-adjoint a , the spectral bound $-\|a\| \mathbf{1} \leq a \leq \|a\| \mathbf{1}$ shows $\mathbf{1}$ is an order unit and that the order-unit norm equals the spectral-radius (operator) norm; the Archimedean property holds because $B(\mathcal{H})_{\text{sa}}$ is finite-dimensional. In the channel setting of this report the order-unit structure of the relevant subspace is inherited *exactly* from $B(\mathcal{H})_{\text{sa}}$; only the product will be approximate.

The single positivity inequality available for merely positive unital maps is Kadison's, with its symmetrized strengthening due to Størmer.

Proposition 2.5 (Kadison and Jordan–Schwarz inequalities). (CITED) *Let $\Phi : B(\mathcal{H}) \rightarrow B(\mathcal{H})$ be a positive unital map.*

(i) (Kadison [5]) *For every self-adjoint $a \in B(\mathcal{H})_{\text{sa}}$,*

$$\Phi(a)^2 \leq \Phi(a^2).$$

(ii) (Jordan–Schwarz, Størmer [7, 8]) *For every $a \in B(\mathcal{H})$, writing $\circ$ for the Jordan product $x \circ y = \frac{1}{2}(xy + yx)$,*

$$\Phi(a^*) \circ \Phi(a) \leq \Phi(a^* \circ a).$$

Part (i) is the case $a = a^$ of part (ii). A merely positive unital map need not satisfy the Schwarz inequality $\Phi(a)^* \Phi(a) \leq \Phi(a^* a)$, which requires 2-positivity.*

Remark 2.6 (Operator norm in place of the cb-norm). (PROVED) (*framing; original to this report*) Kitaev [6] measures non-idempotence in the completely bounded norm, $\|\Phi^2 - \Phi\|_{\text{cb}} \leq \eta$, which presupposes that the relevant maps are completely bounded. We instead use the operator norm of the map on $B(\mathcal{H})_{\text{sa}}$, $\|\Phi^2 - \Phi\| \leq \eta$. The reason is structural: mere positivity is not stable under matrix amplification: $\text{id}_n \otimes \Phi$ need not be positive. In finite dimension every linear map is bounded (and completely bounded once an operator-space structure is fixed), so the obstruction is not existence of a cb-norm; it is that the cb/tensor norm is not the order-norm control supplied by positivity alone. The cb-norm and complete positivity enter Kitaev's argument only through the dilation/tensor machinery (Stinespring, Choi); the operator-norm formulation keeps every estimate intrinsic to $B(\mathcal{H})_{\text{sa}}$ and dilation-free, at the cost of the weaker exponent recorded in the bridge theorem (Section 6). Throughout, the spectral idempotent is denoted P and the projected Jordan product is written $a \bullet b := P(a \circ b)$.

Agent B review correction. The earlier draft said that a positive map need not be completely bounded. In the finite-dimensional setting of this report that sentence is false; the correct issue is failure of positivity under amplification.

3 Jordan algebras and JB-algebras

This section fixes the algebraic vocabulary used throughout the report. Every object here is standard; we follow the axioms in Hanche-Olsen and Størmer [3] so that later approximate versions (Section 5) can be read as quantitative relaxations of these exact identities. We work over a field of characteristic not 2; for the analytic results the scalars are real, as indicated.

3.1 Jordan algebras

The Jordan product symmetrises an associative product: for operators on a Hilbert space, $a \circ b = \frac{1}{2}(ab + ba)$ is self-adjoint whenever a and b are, so it keeps us inside $B(\mathcal{H})_{\text{sa}}$ where the associative product ab does not. Abstracting the two identities that this symmetrised product satisfies gives the notion of a Jordan algebra.

Definition 3.1 (Jordan algebra). (CITED) For elements a, b of an associative algebra, the *Jordan product* is

$$a \circ b = \frac{1}{2}(ab + ba). \quad (1)$$

An algebra A with a product written $(a, b) \mapsto a \circ b$ is called a *Jordan algebra* if the following two identities hold for all $a, b \in A$:

$$a \circ b = b \circ a, \quad (2)$$

$$a \circ (b \circ a^2) = (a \circ b) \circ a^2. \quad (3)$$

Identity (2) states that A is commutative; (3) is the *Jordan identity*. Here $a^2 := a \circ a$.

Remark 3.2 (Form of the Jordan identity). (CITED) The Jordan identity (3), written $a \circ (b \circ a^2) = (a \circ b) \circ a^2$ in [3], is the same as the symmetric-looking form

$$((a \circ a) \circ b) \circ a = (a \circ a) \circ (b \circ a),$$

obtained by writing $a^2 = a \circ a$ and applying commutativity (2) to each product. We use whichever form is more convenient; they are equivalent in any commutative algebra.

A Jordan algebra need not be associative. Its one indispensable substitute for associativity is that any single element, together with its powers, generates an associative subalgebra. This is what makes a functional calculus possible later.

Proposition 3.3 (Power-associativity). (CITED) Let A be a Jordan algebra and $a \in A$. Define powers inductively by $a^1 = a$ and $a^{n+1} = a \circ a^n$ (and $a^0 = \mathbf{1}$ if A is unital). Then the subalgebra generated by a is associative; equivalently, for all natural numbers m, n ,

$$a^{m+n} = a^m \circ a^n. \quad (4)$$

Consequently every power a^n is unambiguously defined.

3.2 JB-algebras

To do analysis one adds a complete norm compatible with the product. The two extra norm conditions below are the Jordan analogue of the C^* -identity: the first fixes the norm of a square, the second forces sums of squares to dominate their summands and is what makes the order structure (positivity = being a square) well-behaved.

Definition 3.4 (JB-algebra). (CITED) A *Jordan Banach algebra* is a real Jordan algebra A equipped with a complete norm satisfying

$$\|a \circ b\| \leq \|a\| \|b\|, \quad a, b \in A. \quad (5)$$

A *JB-algebra* is a Jordan Banach algebra A whose norm satisfies, in addition, the following two conditions for all $a, b \in A$:

$$\|a^2\| = \|a\|^2, \quad (6)$$

$$\|a^2\| \leq \|a^2 + b^2\|. \quad (7)$$

If A is unital with unit $\mathbf{1}$, then (6) gives $\|\mathbf{1}\| = 1$.

The motivating example is concrete: for a complex Hilbert space $\mathcal{H}$, the self-adjoint operators $B(\mathcal{H})_{\text{sa}}$ form a JB-algebra under (1) and the operator norm. A norm-closed Jordan subalgebra of some $B(\mathcal{H})_{\text{sa}}$ is called a *JC-algebra* [3, 3.1.2], and every JC-algebra is a JB-algebra. The content of the finite-dimensional theory is that, with one exception, the converse also holds.

3.3 Finite-dimensional classification

In finite dimensions the JB-algebras are exactly the classical objects of Jordan, von Neumann and Wigner [4]: a real Jordan algebra A is called *formally real* (or Euclidean) if $\sum_{i=1}^n a_i^2 = 0$ in A forces $a_1 = \dots = a_n = 0$. A finite-dimensional unital formally real Jordan algebra is a JB-algebra in its order norm [3, 3.1.7], and conversely every JB-algebra is formally real [3, 3.3.8]; so in finite dimensions the two classes coincide. Their simple pieces are completely classified.

Theorem 3.5 (Jordan–von Neumann–Wigner classification). (CITED) *Every finite-dimensional, formally real, unital Jordan algebra is a direct sum of simple ones. Each simple summand is isomorphic to exactly one of*

- the one-dimensional algebra $\mathbb{R}$;
- a spin factor $\mathbb{R}\mathbf{1} \oplus H$, where H is a real inner-product space of dimension ≥ 2 (the case of two minimal projections summing to $\mathbf{1}$);
- a Hermitian matrix algebra $H_n(\mathbb{R})$, $H_n(\mathbb{C})$, or $H_n(\mathbb{H})$ with $n \geq 3$, where $\mathbb{H}$ denotes the quaternions; or
- the Albert algebra $H_3(\mathbb{O})$ of 3×3 Hermitian matrices over the octonions $\mathbb{O}$.

Of these, only $H_3(\mathbb{O})$ is exceptional: it is not isomorphic to a Jordan subalgebra of self-adjoint operators on any Hilbert space; all the others are JC-algebras.

Agent B review correction. The one-dimensional summand $\mathbb{R}$ is included explicitly; it is the $n = 1$ case in Hanche-Olsen–Størmer Theorem 2.9.6 and is not covered by the listed spin factors.

Remark 3.6 (Status of the exceptional algebra). (CITED) The exceptionality of $H_3(\mathbb{O})$ traces back to the non-associativity of the octonions [3, 2.8.5]. It is the only obstruction to representing a finite-dimensional formally real Jordan algebra by self-adjoint operators, and it is the reason the structure theory of Section 10 must allow for a Jordan, rather than associative, limit object.

4 The spectral idempotent

Throughout this section $\Phi: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ is a fixed unital positive linear map that is *almost idempotent*: there is a parameter $\eta \in [0, \frac{1}{4})$ with

$$\|\Phi^2 - \Phi\| \leq \eta, \quad (8)$$

where $\|\cdot\|$ on the left is the operator norm of a linear map on $B(\mathcal{H})_{\text{sa}}$ (the order-unit norm $\|x\| = \inf\{t > 0 : -t\mathbf{1} \leq x \leq t\mathbf{1}\}$ being used on $B(\mathcal{H})_{\text{sa}}$ itself). The goal is to replace Φ by an honest idempotent P that is close to Φ and inherits its order properties up to an error of size η . The construction is Kitaev's; we record the elementary bounds we need with explicit dependence on η .

We work inside the Banach algebra $\text{End}(B(\mathcal{H})_{\text{sa}})$ of bounded linear operators on $B(\mathcal{H})_{\text{sa}}$, whose unit we also denote $\mathbf{1}$. Since Φ is unital positive it is a contraction for the order-unit norm, so $\|\Phi\| = 1$.

Definition 4.1 (Spectral idempotent, (CITED)). Set $S = 2\Phi - \mathbf{1} \in \text{End}(B(\mathcal{H})_{\text{sa}})$. Define

$$R = (S^2)^{-1/2} \quad (9)$$

by the binomial series for $(\mathbf{1} - (\mathbf{1} - S^2))^{-1/2}$, set $\text{sgn}(S) = SR$, and put

$$P = \theta(2\Phi - \mathbf{1}) = \frac{1}{2}(\mathbf{1} + \text{sgn}(S)) = \frac{1}{2}\left(\mathbf{1} + (2\Phi - \mathbf{1})(\mathbf{1} - 4(\Phi - \Phi^2))^{-1/2}\right). \quad (10)$$

This is the spectral idempotent of Φ for the spectral cluster near 1 [6, Eq. (tilde_Φ), Prop. P].

The series defining R converges because $S^2 - \mathbf{1}$ is small: from (8),

$$\|S\| \leq 2\|\Phi\| + 1 \leq 3, \quad \|S^2 - \mathbf{1}\| = 4\|\Phi^2 - \Phi\| \leq 4\eta < 1, \quad (11)$$

the last inequality holding precisely for $\eta < \frac{1}{4}$. Hence R is a well-defined element of $\text{End}(B(\mathcal{H})_{\text{sa}})$ and is a function of S^2 .

Lemma 4.2 (Properties of P , (PROVED)). ✓ *af-validated* There are universal constants $\eta_0 < \frac{1}{4}$ and C such that, whenever $0 \leq \eta \leq \eta_0$, the operator P of Definition 4.1 satisfies:

- (i) $P^2 = P$ (idempotent);
- (ii) $P(\mathbf{1}) = \mathbf{1}$ (unital);
- (iii) P is a real-linear operator on $B(\mathcal{H})_{\text{sa}}$ (it preserves the self-adjoint part);
- (iv) $\|R - \mathbf{1}\| \leq C\eta$;
- (v) $\delta := \|P - \Phi\| \leq C\eta$;
- (vi) $\|P\| \leq 1 + C\eta$;
- (vii) P is δ -positive: if $a \geq 0$ then $P(a) \geq -\delta\|a\|\mathbf{1}$.

Proof. (i) Because R is a function of S^2 it commutes with S^2 , so $\text{sgn}(S)^2 = S^2 R^2 = S^2 (S^2)^{-1} = \mathbf{1}$. Therefore

$$P^2 = \frac{1}{4}(\mathbf{1} + \text{sgn}(S))^2 = \frac{1}{4}(\mathbf{1} + 2\text{sgn}(S) + \text{sgn}(S)^2) = \frac{1}{2}(\mathbf{1} + \text{sgn}(S)) = P.$$

(ii) Since $\Phi(\mathbf{1}) = \mathbf{1}$ we have $S(\mathbf{1}) = \mathbf{1}$, hence $(S^2 - \mathbf{1})(\mathbf{1}) = 0$. As R is a power series in $S^2 - \mathbf{1}$ with constant term $\mathbf{1}$, this gives $R(\mathbf{1}) = \mathbf{1}$, so $\text{sgn}(S)(\mathbf{1}) = SR(\mathbf{1}) = \mathbf{1}$ and $P(\mathbf{1}) = \frac{1}{2}(\mathbf{1} + \mathbf{1}) = \mathbf{1}$.

(iii) The construction takes place in $\text{End}(B(\mathcal{H})_{\text{sa}})$, and every term in the binomial series is a real-linear endomorphism of $B(\mathcal{H})_{\text{sa}}$. Hence R , $\text{sgn}(S) = SR$, and $P = \frac{1}{2}(\mathbf{1} + \text{sgn}(S))$ all map $B(\mathcal{H})_{\text{sa}}$ to itself.

(iv) Writing $u = \mathbf{1} - S^2$ with $\|u\| \leq 4\eta$ from (11), the binomial series gives $R = \sum_{n \geq 0} \binom{-1/2}{n} (-u)^n$, so

$$\|R - \mathbf{1}\| \leq \sum_{n \geq 1} \left| \binom{-1/2}{n} \right| \|u\|^n \leq \frac{\|u\|/2}{1 - \|u\|} \leq \frac{2\eta}{1 - 4\eta}.$$

Choosing $\eta_0 \leq \frac{1}{8}$ gives $\|R - \mathbf{1}\| \leq 4\eta \leq C\eta$. Here the coefficients satisfy $|\binom{-1/2}{n}| \leq \frac{1}{2}$ for $n \geq 1$, so the tail is dominated by the geometric series $\frac{1}{2} \sum_{n \geq 1} \|u\|^n$.

(v) Since $P - \Phi = \frac{1}{2}(\text{sgn}(S) - S) = \frac{1}{2}S(R - \mathbf{1})$,

$$\delta = \|P - \Phi\| = \frac{1}{2}\|S(R - \mathbf{1})\| \leq \frac{1}{2}\|S\| \|R - \mathbf{1}\| \leq \frac{3}{2}C\eta \leq C\eta,$$

using $\|S\| \leq 3$ and (iv).

(vi) By the triangle inequality and (v), $\|P\| \leq \|\Phi\| + \delta \leq 1 + C\eta$.

(vii) Let $a \geq 0$. Then $\Phi(a) \geq 0$ because Φ is positive, and $\|P(a) - \Phi(a)\| \leq \delta\|a\|$ by (v). For self-adjoint elements, $\|x - y\| \leq \beta$ forces $x \geq y - \beta\mathbf{1}$; applied with $x = P(a)$, $y = \Phi(a) \geq 0$, and $\beta = \delta\|a\|$, this yields $P(a) \geq \Phi(a) - \delta\|a\|\mathbf{1} \geq -\delta\|a\|\mathbf{1}$. $\square$

Remark 4.3 (Kitaev comparison). (CITED; PROVED ADAPTATION) The same operator was introduced by Kitaev for unital completely positive maps, using the completely bounded norm $\|\cdot\|_{\text{cb}}$ in place of the operator norm in (8); the algebraic identities (i)–(ii) and the bounds (iv)–(vi) are norm-independent, so they apply here [6, Prop. P]. The operator-norm hypothesis used above is what Lemma 4.2 (vii) requires, and it is all that the bridge of the following sections needs.

By Lemma 4.2, P is an idempotent, unital real-linear operator on $B(\mathcal{H})_{\text{sa}}$ with $\delta = \|P - \Phi\| \leq C\eta$. Its range $A = \text{Im } P = \text{Ker}(\mathbf{1} - P)$ is a real subspace of $B(\mathcal{H})_{\text{sa}}$ containing $\mathbf{1}$, and the *projected Jordan product*

$$a \bullet b := P(a \circ b), \quad a, b \in A, \tag{12}$$

will be analysed in Section 5.

Agent B review correction. This section works on the real self-adjoint space $B(\mathcal{H})_{\text{sa}}$. The previous “ $\dagger$ -preserving” formulation was inherited from the complex $B(\mathcal{H})$ statement in Kitaev and was stronger than the domain used here; the bridge proof only needs preservation of $B(\mathcal{H})_{\text{sa}}$.

5 Approximate Jordan algebras

We now introduce the central object of this work: a finite-dimensional real order-unit space carrying a commutative bilinear product that satisfies the JB-algebra axioms of Section 3 only up to a controlled error ε . The structural choice, motivated by the spectral construction of Section 4 (where such an object arises from an almost-idempotent positive map), is that the *order and norm are exact* and only the *product* is approximate. This is the project’s working definition.

Definition 5.1 (ε -JB algebra, order-unit form). (CONSENSUS) Let $\varepsilon \geq 0$. An ε -JB algebra (order-unit form) is a finite-dimensional real order-unit space $(A, \mathbf{1}, \leq)$ with order-unit norm $\|\cdot\|$, together with a commutative bilinear product $\circ: A \times A \rightarrow A$ for which commutativity, $a \circ b = b \circ a$, and the unit law, $\mathbf{1} \circ a = a$, hold *exactly*, and such that for all $a, b \in A$:

$$\|a \circ b\| \leq (1 + \varepsilon) \|a\| \|b\|, \tag{JB1}$$

$$\|a \circ a\| \geq (1 - \varepsilon) \|a\|^2, \tag{JB2}$$

$$a \circ a \geq -\varepsilon \|a\|^2 \mathbf{1}, \tag{JB3}$$

$$\|((a \circ a) \circ b) \circ a - (a \circ a) \circ (b \circ a)\| \leq \varepsilon \|a\|^3 \|b\|. \tag{JB4}$$

We call (JB1) *approximate submultiplicativity*, (JB2) the *lower square norm bound*, (JB3) *approximate positivity of squares*, and (JB4) the *approximate Jordan identity*. The status (CONSENSUS) records that this is the definition agreed between the authors; that it is actually *realised*, with $\varepsilon \leq C\sqrt{\eta}$, by the spectral construction of Section 4 is the separate (PROVED) content of Theorem 6.1.

Remark 5.2. (PROVED FROM DEFINITIONS) At $\varepsilon = 0$ each axiom degenerates to a genuine JB-algebra axiom of Section 3: (JB1) becomes submultiplicativity $\|a \circ b\| \leq \|a\| \|b\|$; (JB2) forces $\|a \circ a\| = \|a\|^2$ (its upper bound is the $\varepsilon = 0$ case of (JB1)), the defining JB norm condition on squares; (JB3) becomes $a \circ a \geq 0$, the formally-real (order) axiom that squares are positive; and (JB4) becomes the exact Jordan identity $(a^2 \circ b) \circ a = a^2 \circ (b \circ a)$ with $a^2 := a \circ a$. Thus an ε -JB algebra is exactly a quantitative relaxation of a JB-algebra in the product alone. The order relation $\leq$, the unit $\mathbf{1}$, the order-unit norm $\|\cdot\|$, and the laws $a \circ b = b \circ a$ and $\mathbf{1} \circ a = a$ are kept *exact*. This is the key structural simplification relative to Kitaev’s ε - C^* algebras [6], where the norm itself is only approximately multiplicative and the unit may be approximate (e.g. his $\|XY\| \leq (1 + \varepsilon)\|X\|\|Y\|$, $\|X^\dagger X\| \geq (1 - \varepsilon)\|X\|^2$, $\|I - 1\| \leq \varepsilon$); here only $\circ$ carries the defect, which removes the approximate-norm bookkeeping at the cost of nothing in the intended application.

Definition 5.3 (δ -Jordan-homomorphism, δ -isomorphism). (CONSENSUS) Let A, B be ε -JB algebras (Definition 5.1) and $\delta \geq 0$. A δ -Jordan-homomorphism is a linear map $v: A \rightarrow B$ that almost preserves the unit and the product:

$$\|v(\mathbf{1}) - \mathbf{1}\| \leq \delta, \quad (13)$$

$$\|v(a \circ b) - v(a) \circ v(b)\| \leq \delta \|a\| \|b\| \quad (a, b \in A). \quad (14)$$

A δ -isomorphism is a bijective δ -Jordan-homomorphism that is moreover an approximate isometry,

$$(1 - \delta) \|a\| \leq \|v(a)\| \leq (1 + \delta) \|a\| \quad (a \in A). \quad (15)$$

This is an algebraic and normed notion of approximate Jordan isomorphism. It does not, by itself, say that v or v^{-1} approximately preserves the positive cone; that order control is an additional requirement in the positive-map factorization problem. The status (CONSENSUS) marks this as the authors’ adopted definition — the Jordan (commutative-product) transcription of Kitaev’s δ -homomorphism and δ -isomorphism [6].

Agent B review correction. The earlier draft called this equivalently an approximate order-isomorphism. That was not justified by the displayed hypotheses and has been separated from the normed algebraic definition.

6 The bridge theorem

This section contains the central result of the report: from a unital positive almost-idempotent map Φ one constructs, with no use of dilations and no appeal to complete positivity, a space on which the Jordan-algebra axioms hold up to an error $O(\sqrt{\eta})$. The construction and every estimate below are transcribed from the verified internal proof, with notation harmonised to the conventions of this report. The exact (non-perturbative) analogue—that a positive projection onto a unital subspace endows its range with a genuine Jordan structure—is due to Effros and Størmer [2]; the present theorem is the quantitative, almost-idempotent version of that statement.

Throughout, $\mathcal{H}$ is finite-dimensional, $V = B(\mathcal{H})_{\text{sa}}$ carries the order-unit (operator) norm, and the ambient Jordan product is

$$x \circ y = \frac{1}{2}(xy + yx), \quad x, y \in V.$$

We write $a \bullet b := P(a \circ b)$ for the projected Jordan product, where P is the spectral idempotent of Section 4. **All constants C below are universal: they depend on nothing, and in particular are independent of $\dim \mathcal{H}$ (dimension-free).** We are free to decrease the threshold η_0 whenever convenient, in order to absorb an $O(\eta)$ term into an $O(\sqrt{\eta})$ term.

Theorem 6.1 (Algebraic bridge). (PROVED) ✓ af-validated *There are universal constants $\eta_0 > 0$ and C with the following property. Let $\mathcal{H}$ be finite-dimensional, let $V = B(\mathcal{H})_{\text{sa}}$ with the order-unit/operator norm and Jordan product $x \circ y = \frac{1}{2}(xy + yx)$, and let $\Phi: V \rightarrow V$ be unital and positive with*

$$\|\Phi^2 - \Phi\| \leq \eta \leq \eta_0.$$

Let $P = \theta(2\Phi - 1)$ be the spectral idempotent associated to the spectral cluster of Φ near 1 (Section 4), and set

$$A = \text{Im } P, \quad A_+ = A \cap V_+, \quad a \bullet b = P(a \circ b) \quad (a, b \in A).$$

Then $(A, \bullet, \mathbf{1}, A_+)$, with the order structure inherited from V , is an ε -JB order-unit algebra (Section 5) with

$$\varepsilon \leq C\sqrt{\eta}.$$

In particular, for all $a, b \in A$,

$$\mathbf{1} \bullet a = a, \tag{16}$$

$$\|a \bullet b\| \leq (1 + C\eta) \|a\| \|b\|, \tag{17}$$

$$a \bullet a \geq -C\eta \|a\|^2 \mathbf{1}, \tag{18}$$

$$\|a \bullet a\| \geq (1 - C\eta) \|a\|^2, \tag{19}$$

$$\|((a \bullet a) \bullet b) \bullet a - (a \bullet a) \bullet (b \bullet a)\| \leq C\sqrt{\eta} \|a\|^3 \|b\|. \tag{20}$$

Spectral estimate. We first record the basic estimates on P established in Section 4; they are used repeatedly below. Write

$$\delta = \|P - \Phi\|.$$

Since Φ is unital and positive, it is a contraction for the order-unit norm, $\|\Phi\| = 1$. Setting $S = 2\Phi - \mathbf{1}$ in the Banach algebra $\text{End}(V)$ one has

$$\|S\| \leq 3, \quad \|S^2 - \mathbf{1}\| = 4\|\Phi^2 - \Phi\| \leq 4\eta,$$

and $P = \frac{1}{2}(\mathbf{1} + \text{sgn}(S))$ with $\text{sgn}(S) = S(S^2)^{-1/2}$ defined by the convergent binomial series for η_0 small. This P agrees with $\theta(2\Phi - \mathbf{1})$, satisfies $P^2 = P$, fixes the unit, $P(\mathbf{1}) = \mathbf{1}$, and obeys

$$\delta = \|P - \Phi\| \leq C\eta, \quad \|P\| \leq 1 + C\eta. \tag{21}$$

Moreover P is δ -positive:

$$x \geq 0 \implies P(x) \geq -\delta \|x\| \mathbf{1}, \tag{22}$$

because $\Phi(x) \geq 0$ and $\|P(x) - \Phi(x)\| \leq \delta \|x\|$.

Lemma 0: exact order-unit structure on A

Lemma 6.2 (Exact order unit). (PROVED) ✓ *af-validated* The order-unit part of the structure on A is exact. Since $P(\mathbf{1}) = \mathbf{1}$, the range $A = \text{Im } P$ is a real linear subspace of V containing $\mathbf{1}$; equip it with the inherited cone $A_+ = A \cap V_+$. Then $(A, A_+, \mathbf{1})$ is an order-unit space, and its order-unit norm coincides with the ambient operator norm:

$$\|a\|_{\text{ou}, A} = \inf\{t > 0 : -t\mathbf{1} \leq a \leq t\mathbf{1}\} = \|a\|_{B(\mathcal{H})} \quad (a \in A).$$

Proof. For every $a \in A$ one has $-\|a\| \mathbf{1} \leq a \leq \|a\| \mathbf{1}$ in V , so $\mathbf{1}$ is an order unit for A . The cone is Archimedean because V_+ is closed: if $a + \varepsilon \mathbf{1} \in A_+$ for every $\varepsilon > 0$, then $a \geq 0$ in V , hence $a \in A_+$. The displayed equality of norms is the definition of the order-unit norm together with the fact that, for self-adjoint matrices, the smallest t with $-t\mathbf{1} \leq a \leq t\mathbf{1}$ equals the operator norm. Thus no approximate norm or order structure is introduced; all approximate estimates below concern only the product $\bullet$. $\square$

Standard order estimates

We use only the following order facts, none of which requires complete positivity. Items (O1) and (O2) are the Kadison/Jordan–Schwarz inequalities recalled in Section 2 [5, 8].

- (O1) *Jordan–Schwarz* for unital positive maps on self-adjoint elements: $\Phi(x)^2 \leq \Phi(x^2)$ for $x = x^*$.
- (O2) *Cauchy–Schwarz* for every positive functional ω on $B(\mathcal{H})_{\text{sa}}$: $|\omega(x \circ y)|^2 \leq \omega(x^2) \omega(y^2)$.
- (O3) *Norm monotonicity in the cone*: if $0 \leq x \leq y$ then $\|x\| \leq \|y\|$.
- (O4) *Norm as a supremum over states*: for self-adjoint x , $\|x\| = \sup_{\rho} |\rho(x)|$, the supremum over states ρ on $B(\mathcal{H})$.
- (O5) *Order perturbation*: if x, y are self-adjoint and $\|x - y\| \leq \varepsilon$, then $y - \varepsilon \mathbf{1} \leq x \leq y + \varepsilon \mathbf{1}$.

Lemma 1: the easy axioms

Lemma 6.3 (Easy JB axioms). (PROVED) ✓ *af-validated* For all $a, b \in A$ the identities (16)–(19) hold; that is,

$$\begin{aligned} \mathbf{1} \bullet a &= a, & a \bullet b &= b \bullet a, & \|a \bullet b\| &\leq (1 + C\eta) \|a\| \|b\|, \\ a \bullet a &\geq -C\eta \|a\|^2 \mathbf{1}, & \|a \bullet a\| &\geq (1 - C\eta) \|a\|^2. \end{aligned}$$

Proof. The unit law $\mathbf{1} \bullet a = P(\mathbf{1} \circ a) = P(a) = a$ and commutativity $a \bullet b = P(a \circ b) = P(b \circ a) = b \bullet a$ are exact, the latter because $\circ$ is commutative. The product bound follows from $\|P\| \leq 1 + C\eta$ in (21) and $\|a \circ b\| \leq \|a\| \|b\|$.

For squares, $a \bullet a = P(a^2)$, and applying (22) to $a^2 \geq 0$ gives $a \bullet a \geq -\delta \|a\|^2 \mathbf{1} \geq -C\eta \|a\|^2 \mathbf{1}$.

For the norm lower bound, since $a = P(a)$ we have $\|\Phi(a) - a\| = \|(\Phi - P)(a)\| \leq \delta \|a\|$. Jordan–Schwarz (O1) gives $\Phi(a^2) \geq \Phi(a)^2$, hence by monotonicity (O3)

$$\|\Phi(a^2)\| \geq \|\Phi(a)^2\| = \|\Phi(a)\|^2 \geq (1 - C\delta) \|a\|^2.$$

Since $\|P(a^2) - \Phi(a^2)\| \leq \delta \|a\|^2$, we conclude

$$\|a \bullet a\| = \|P(a^2)\| \geq (1 - C\eta) \|a\|^2.$$

Thus only the Jordan identity (20) remains to be proved. $\square$

Lemma 2: the first insertion estimate

For every state ρ on $B(\mathcal{H})$ set

$$\omega = \rho \circ \Phi, \quad \|x\|_\omega^2 = \omega(x^2). \quad (23)$$

Since Φ is positive, ω is a positive functional and $\|\cdot\|_\omega$ is a seminorm on V .

Lemma 6.4 (First insertion estimate). (PROVED) ✓ af-validated (i) (Almost-contraction.) *For all $x \in V$,*

$$\|Px\|_\omega^2 \leq \|x\|_\omega^2 + C\eta \|x\|^2. \quad (24)$$

(ii) (Almost-orthogonality.) *For $u \in \text{Im } P$ and $n \in \text{Ker } P$,*

$$|\omega(u \circ n)| \leq C\sqrt{\eta} \|u\| \|n\|. \quad (25)$$

(iii) (First insertion (FI).) *For all $x \in V$ and $b \in A$,*

$$\|P(Px \circ b) - P(x \circ b)\| \leq C\sqrt{\eta} \|x\| \|b\| \quad (\text{FI}). \quad (26)$$

Proof. (i) We chain three inequalities:

$$\rho \Phi((Px)^2) \leq \rho \Phi((\Phi x)^2) + C\eta \|x\|^2 \leq \rho \Phi^2(x^2) + C\eta \|x\|^2 \leq \rho \Phi(x^2) + C\eta \|x\|^2.$$

The first inequality uses $\|Px - \Phi x\| \leq \delta \|x\|$ together with the elementary bound $\|y^2 - z^2\| \leq 2 \max(\|y\|, \|z\|) \|y - z\|$ and $\|P\|, \|\Phi\| \leq C$: applying the state $\rho \circ \Phi$ to the self-adjoint difference of the two squares costs at most its operator norm, which is $O(\eta)\|x\|^2$. The middle inequality is Jordan–Schwarz (O1) for Φ applied to Φx ; the last uses $\|\Phi^2 - \Phi\| \leq \eta$. This is (24).

(ii) Since P is an exact idempotent, (24) forces almost-orthogonality of $\text{Im } P$ and $\text{Ker } P$ in every seminorm $\|\cdot\|_\omega$. Put $c = \omega(u \circ n)$ and $a = \|n\|_\omega^2$. Because $P(u + tn) = u$ for every real t , (24) applied to $x = u + tn$ gives

$$\|u\|_\omega^2 \leq \|u + tn\|_\omega^2 + K\eta \|u + tn\|^2 = \|u\|_\omega^2 + 2tc + t^2a + K\eta \|u + tn\|^2,$$

where K is the constant of (24). Choose the sign of t opposite to the sign of c and write $s = |t|$; using $\|u + tn\|^2 \leq 2\|u\|^2 + 2s^2\|n\|^2$ we obtain, for all $s > 0$,

$$2s|c| \leq s^2(a + 2K\eta \|n\|^2) + 2K\eta \|u\|^2.$$

Optimising over $s > 0$ yields

$$|c| \leq C\sqrt{\eta} \|u\| \sqrt{a + \eta \|n\|^2} \leq C\sqrt{\eta} \|u\| \|n\|,$$

the last step using $a = \|n\|_\omega^2 \leq \|n\|^2$. This is (25).

(iii) Let $x \in V$, $b \in A$, and set $n = x - Px \in \text{Ker } P$, so $\|n\| \leq C\|x\|$. For every state ρ , with $\omega = \rho \circ \Phi$,

$$|\rho P(n \circ b)| \leq |\rho \Phi(n \circ b)| + C\eta \|n\| \|b\| = |\omega(n \circ b)| + C\eta \|n\| \|b\| \leq C\sqrt{\eta} \|x\| \|b\|,$$

where the first step replaces P by Φ at cost $\delta \|n \circ b\| \leq C\eta \|n\| \|b\|$ and the last applies (25) with $u = b \in \text{Im } P$. Since $P(n \circ b)$ is self-adjoint and $P(x \circ b) - P(Px \circ b) = P(n \circ b)$, taking the supremum over states (O4) gives (26). $\square$

Lemma 3: square holes are approximately null

For $r \in A$ define the *square hole*

$$q_r = P(r^2) - r^2 \in \text{Ker } P,$$

the inclusion holding because $P(q_r) = P^2(r^2) - P(r^2) = 0$.

Lemma 6.5 (Square holes are almost null). (PROVED) ✓ *af-validated* For every $r \in A$ and every state seminorm $\omega = \rho \circ \Phi$,

$$q_r \geq -C\eta \|r\|^2 \mathbf{1}, \quad (27)$$

$$\|P(q_r^2)\| \leq C\eta \|r\|^4, \quad (28)$$

$$\|q_r\|_\omega \leq C\sqrt{\eta} \|r\|^2. \quad (29)$$

Proof. Since $\|\Phi(r) - r\| \leq \delta \|r\|$, Jordan–Schwarz (O1) and order perturbation (O5) give

$$\Phi(r^2) \geq \Phi(r)^2 \geq r^2 - C\delta \|r\|^2 \mathbf{1},$$

the second inequality from $\|\Phi(r)^2 - r^2\| \leq C\delta \|r\|^2$. Replacing $\Phi(r^2)$ by $P(r^2)$ costs another $\delta \|r\|^2$ in the same order sense, which yields $P(r^2) \geq r^2 - C\delta \|r\|^2 \mathbf{1}$, i.e. (27) (recall $\delta \leq C\eta$).

Choose $\gamma = C\delta \|r\|^2$ so that $y = q_r + \gamma \mathbf{1} \geq 0$. Also $\|q_r\| \leq C\|r\|^2$ and hence $\|y\| \leq C\|r\|^2$. Since $P(q_r) = 0$,

$$\|\Phi(q_r)\| = \|(\Phi - P)(q_r)\| \leq C\delta \|r\|^2,$$

so $0 \leq \Phi(y)$ with $\|\Phi(y)\| \leq C\delta \|r\|^2$. Because $0 \leq y \leq C\|r\|^2 \mathbf{1}$, functional calculus gives $0 \leq y^2 \leq C\|r\|^2 y$, and positivity of Φ together with monotonicity (O3) yields

$$\|\Phi(y^2)\| \leq C\|r\|^2 \|\Phi(y)\| \leq C\delta \|r\|^4.$$

Finally $q_r^2 \leq 2y^2 + 2\gamma^2 \mathbf{1}$, so $\|\Phi(q_r^2)\| \leq C\delta \|r\|^4$; the term $\gamma^2 \mathbf{1}$ is $O(\delta^2 \|r\|^4)$ and is absorbed into $O(\delta \|r\|^4)$ after decreasing η_0 . Replacing Φ by P costs at most $\delta \|q_r^2\| \leq C\delta \|r\|^4$, proving (28). Estimate (29) follows since $\|q_r\|_\omega^2 = \rho \Phi(q_r^2) \leq \|\Phi(q_r^2)\| \leq C\eta \|r\|^4$. $\square$

Lemma 4: polarised hole estimates

For $r, s \in A$ set the *product hole*

$$q_{r,s} = P(r \circ s) - r \circ s = -h_{r,s},$$

a symmetric bilinear function of (r, s) lying in $\text{Ker } P$.

Lemma 6.6 (Polarised hole estimates). (PROVED) ✓ *af-validated* For all $r, s, u, v \in A$, all $z \in V$, and every state seminorm $\omega = \rho \circ \Phi$,

$$\|q_{r,s}\|_\omega \leq C\sqrt{\eta} \|r\| \|s\|, \quad (30)$$

$$\|q_{r,s}\| \leq C\|r\| \|s\|, \quad (31)$$

$$\|P(q_{r,s} \circ q_{u,v})\| \leq C\eta \|r\| \|s\| \|u\| \|v\| \quad (\text{HH}), \quad (32)$$

$$\|P(q_{r,s} \circ z)\| \leq C\sqrt{\eta} \|r\| \|s\| \|z\| \quad (\text{HZ}). \quad (33)$$

Proof. Estimate (30). For $\lambda > 0$, polarisation of the symmetric bilinear map $(r, s) \mapsto q_{r,s}$ gives

$$\lambda q_{r,s} = q_{\lambda r,s} = \frac{1}{4}(q_{\lambda r+s} - q_{\lambda r-s}),$$

where $q_x = P(x^2) - x^2$ is the square hole of Lemma 6.5. Hence (29) gives

$$\|q_{r,s}\|_\omega \leq C\sqrt{\eta}(\lambda\|r\|^2 + \lambda^{-1}\|s\|^2).$$

If $r = 0$ or $s = 0$ this is trivial; otherwise put $\lambda = \|s\|/\|r\|$ to get (30). The crude bound (31) is immediate from $\|P\| \leq C$ and $\|r \circ s\| \leq \|r\| \|s\|$.

Estimate (32). For every state ρ , Cauchy–Schwarz (O2) for the positive functional $\omega = \rho \circ \Phi$ gives

$$|\rho \Phi(q_{r,s} \circ q_{u,v})| \leq \|q_{r,s}\|_\omega \|q_{u,v}\|_\omega \leq C\eta \|r\| \|s\| \|u\| \|v\|$$

by (30). Replacing Φ by P costs at most $\delta \|q_{r,s}\| \|q_{u,v}\|$, which has the same bound after increasing C (using (31)). Since $P(q_{r,s} \circ q_{u,v})$ is self-adjoint, the state supremum (O4) is its operator norm.

Estimate (33). For $z \in V$ we have $\|z\|_\omega \leq \|z\|$, because $z^2 \leq \|z\|^2 \mathbf{1}$ and Φ is positive unital. Hence, as above, for every state ρ ,

$$|\rho \Phi(q_{r,s} \circ z)| \leq \|q_{r,s}\|_\omega \|z\|_\omega \leq C\sqrt{\eta} \|r\| \|s\| \|z\|.$$

Replacing Φ by P costs at most $\delta \|q_{r,s}\| \|z\| \leq C\eta \|r\| \|s\| \|z\|$, which is absorbed into the displayed $O(\sqrt{\eta})$ bound. Taking the state supremum (O4) gives (33). $\square$

Lemma 5: one-hole contexts

Lemma 6.7 (One-hole contexts). (PROVED) ✓ af-validated Let $h = h_{r,s} = r \circ s - P(r \circ s)$ be a range-product hole, $r, s \in A$, and let $t, u \in A$. Then

$$\|P((h \circ t) \circ u)\| \leq C\sqrt{\eta} \|r\| \|s\| \|t\| \|u\| \quad (5.1), \quad (34)$$

$$\|P(h \circ (t \circ u))\| \leq C\sqrt{\eta} \|r\| \|s\| \|t\| \|u\| \quad (5.2). \quad (35)$$

Proof. Estimate (34). Put $y = h \circ t$. Since $h = -q_{r,s}$, (33) (with $z = t$) gives $\|P(y)\| \leq C\sqrt{\eta} \|r\| \|s\| \|t\|$, hence

$$\|P(P(y) \circ u)\| \leq \|P\| \|P(y)\| \|u\| \leq C\sqrt{\eta} \|r\| \|s\| \|t\| \|u\|.$$

For the kernel part $y - P(y)$, apply (26) with $x = y$ and $b = u$, using the crude bound $\|y\| \leq C\|r\| \|s\| \|t\|$:

$$P((y - P(y)) \circ u) = P(y \circ u) - P(P(y) \circ u)$$

has the same $O(\sqrt{\eta} \|r\| \|s\| \|t\| \|u\|)$ bound. Adding the two parts gives (34).

Estimate (35). Write $t \circ u = P(t \circ u) + h_{t,u}$. The first term gives $P(h \circ P(t \circ u))$, controlled by (33) because $P(t \circ u) \in A$ and $h = -q_{r,s}$; the second gives $P(h \circ h_{t,u}) = P(q_{r,s} \circ q_{t,u})$, controlled by (32), hence $O(\eta \|r\| \|s\| \|t\| \|u\|)$ and absorbed into the $O(\sqrt{\eta})$ target. $\square$

Assembly: the Jordan identity

We now prove (20), the last remaining axiom. The strategy is to push both sides through the exact Jordan identity of the ambient product $\circ$ recalled in Section 3, paying only $O(\sqrt{\eta})$ for each hole created in the process.

Proposition 6.8 (Approximate Jordan identity). (PROVED) ✓ af-validated For all $a, b \in A$,

$$\|((a \bullet a) \bullet b) \bullet a - (a \bullet a) \bullet (b \bullet a)\| \leq C\sqrt{\eta} \|a\|^3 \|b\|.$$

Proof. Fix $a, b \in A$ and write

$$p = P(a^2) = a^2 - h_{a,a}, \quad q = P(b \circ a) = b \circ a - h_{b,a},$$

so that $p, q \in A$ and, by the crude bounds,

$$\|p\| \leq C\|a\|^2, \quad \|q\| \leq C\|a\| \|b\|.$$

The left side. Let $l = h_{p,b} = p \circ b - P(p \circ b)$. Then

$$((a \bullet a) \bullet b) \bullet a = P(P(p \circ b) \circ a) = P((p \circ b) \circ a) - P(l \circ a).$$

Here $l \in \text{Ker } P$, since $P(l) = P(p \circ b) - P^2(p \circ b) = 0$, and

$$\|l\| \leq \|p \circ b\| + \|P(p \circ b)\| \leq C\|a\|^2 \|b\|.$$

As $P(l) = 0$, (26) applied with $x = l$ and range element a gives

$$\|P(l \circ a)\| \leq C\sqrt{\eta} \|a\|^3 \|b\| \quad (6.1). \quad (36)$$

Next, using $p = a^2 - h_{a,a}$,

$$P((p \circ b) \circ a) = P((a^2 \circ b) \circ a) - P((h_{a,a} \circ b) \circ a),$$

and the last term is $O(\sqrt{\eta} \|a\|^3 \|b\|)$ by (34) with $h = h_{a,a}$, $t = b$, $u = a$. Combining with (36),

$$((a \bullet a) \bullet b) \bullet a = P((a^2 \circ b) \circ a) + O(\sqrt{\eta} \|a\|^3 \|b\|) \quad (\text{L}). \quad (37)$$

The right side. Expanding both holes,

$$(a \bullet a) \bullet (b \bullet a) = P(p \circ q) = P((a^2 - h_{a,a}) \circ (b \circ a - h_{b,a})),$$

which equals

$$P(a^2 \circ (b \circ a)) - P(h_{a,a} \circ (b \circ a)) - P(a^2 \circ h_{b,a}) + P(h_{a,a} \circ h_{b,a}).$$

We bound the three error terms.

- $\|P(h_{a,a} \circ (b \circ a))\| \leq C\sqrt{\eta} \|a\|^3 \|b\|$ by (35) with $h = h_{a,a}$, $t = b$, $u = a$.
- For the term $P(a^2 \circ h_{b,a})$, decompose $a^2 = p + h_{a,a}$:

$$P(a^2 \circ h_{b,a}) = P(p \circ h_{b,a}) + P(h_{a,a} \circ h_{b,a}).$$

Since $h_{b,a} \in \text{Ker } P$ and $p \in A$, (26) applied with $x = h_{b,a}$ and range element p gives $P(P(h_{b,a}) \circ p) = 0$, whence

$$\|P(p \circ h_{b,a})\| \leq C\sqrt{\eta} \|p\| \|h_{b,a}\| \leq C\sqrt{\eta} \|a\|^3 \|b\|.$$

- Both occurrences of $P(h_{a,a} \circ h_{b,a}) = P(q_{a,a} \circ q_{b,a})$ are bounded by (32): $\|P(h_{a,a} \circ h_{b,a})\| \leq C\eta \|a\|^3 \|b\|$, which for $\eta \leq \eta_0$ is absorbed into the $O(\sqrt{\eta})$ error.

Therefore

$$(a \bullet a) \bullet (b \bullet a) = P(a^2 \circ (b \circ a)) + O(\sqrt{\eta} \|a\|^3 \|b\|) \quad (\text{R}). \quad (38)$$

Cancellation of the leading term. The ambient product $\circ$ satisfies the *exact* Jordan identity (Section 3):

$$(a^2 \circ b) \circ a = a^2 \circ (b \circ a).$$

Hence the leading terms $P((a^2 \circ b) \circ a)$ in (37) and $P(a^2 \circ (b \circ a))$ in (38) are identical. Subtracting (38) from (37) leaves only the error terms:

$$\|((a \bullet a) \bullet b) \bullet a - (a \bullet a) \bullet (b \bullet a)\| \leq C\sqrt{\eta} \|a\|^3 \|b\|. \quad \square$$

Proof of Theorem 6.1. Lemma 6.2 establishes that $(A, A_+, \mathbf{1})$ is an order-unit space with order-unit norm equal to the ambient operator norm, so the order structure is exact. Lemma 6.3 gives the unit law, commutativity, the product bound (17), the square positivity (18), and the square-norm lower bound (19). Proposition 6.8 gives the approximate Jordan identity (20). Together these are exactly the axioms of an ε -JB order-unit algebra (Section 5) with $\varepsilon \leq C\sqrt{\eta}$. All constants are universal and dimension-free. $\square$

Remark 6.9. (CITED) The non-perturbative limit $\eta = 0$ recovers the exact statement of Effros and Størmer [2]: a positive unital projection P on a unital JC-algebra makes its range a JC-algebra under $a \bullet b = P(a \circ b)$. Theorem 6.1 is the quantitative, almost-idempotent version, in which every hole estimate degrades gracefully as $O(\sqrt{\eta})$ or $O(\eta)$ and the exact Jordan identity of the ambient algebra supplies the leading-order cancellation.

Remark 6.10 (Numerical corroboration; the null square is linear in η). (NUMERICAL) The square-hole estimate (28) is genuinely linear in η , not better: in an explicit classical (commutative) $\mathbb{R}^3$ family of almost-idempotent stochastic maps one finds $\|P(q_r^2)\|/\eta \rightarrow 32/27$ as $\eta \rightarrow 0$, so $O(\eta^2)$ is false and the exponent in (28) is sharp. The same probe shows the qubit case is trivial: for $\mathcal{H} = \mathbb{C}^2$ the range $\text{Im } P$ is closed under the ambient Jordan product, so all holes $h_{r,s}$ vanish and no information about the Jordan defect is visible below $\dim \mathcal{H} = 3$. These are numerical observations on small examples, consistent with — but not a substitute for — the proved bounds above.

7 Faithful invariant states and the ambient product

The bridge of Section 6 makes $A = \text{Im } P$ an approximate Jordan algebra under the *projected* product $a \bullet b = P(a \circ b)$. The exact fixed-point theory has a sharper companion: when the positive projection admits a *faithful invariant state*, its range is already closed under the *ambient* Jordan product $\circ$ of $B(\mathcal{H})_{\text{sa}}$, so that $\bullet$ and $\circ$ coincide on the range and no projection is needed. This is the multiplicative-domain mechanism used in the faithful fixed-point results of van Luijk–Wilming. This section asks whether that sharper, ambient-product statement survives to the almost-idempotent setting. The answer is instructive and was sharpened in adversarial exchange between the two authors: **the transfer is governed by the quantitative conditioning of the invariant state, and the bare hypothesis “admits a faithful invariant state” is not sufficient.**

Throughout, $\Phi: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ is unital and positive, $\|\Phi^2 - \Phi\| \leq \eta \leq \eta_0$, $P = \theta(2\Phi - \mathbf{1})$, $A = \text{Im } P$, and for $a, b \in A$ we keep the bridge’s holes

$$q_a = P(a^2) - a^2 \in \text{Ker } P, \quad h_{a,b} = a \circ b - P(a \circ b) = -q_{a,b} \in \text{Ker } P.$$

A state ω on $B(\mathcal{H})$ has a density ρ_ω with $\omega(x) = \text{Tr}(\rho_\omega x)$; it is *faithful* iff $\rho_\omega > 0$, and we write $\lambda_\omega = \lambda_{\min}(\rho_\omega) > 0$ for its least eigenvalue (its *conditioning*).

The exact statement

Proposition 7.1 (Faithful invariance forces the ambient product). (PROVED) *Let $P: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ be a unital positive idempotent ($P^2 = P$) admitting a faithful state ω with $\omega \circ P = \omega$. Then $A = \text{Im } P$ is closed under the ambient Jordan product: $P(a \circ b) = a \circ b$ for all $a, b \in A$. Thus $\bullet$ and $\circ$ agree on A , and A is a unital Jordan subalgebra of $(B(\mathcal{H})_{\text{sa}}, \circ)$.*

Proof. For $a \in A$ we have $P(a) = a$, so Kadison's inequality (the self-adjoint Jordan–Schwarz case, Section 2) gives $P(a)^2 \leq P(a^2)$, i.e. $q_a = P(a^2) - a^2 \geq 0$. Invariance gives $\omega(q_a) = \omega(P(a^2)) - \omega(a^2) = \omega(a^2) - \omega(a^2) = 0$. A faithful state is faithful on the positive cone (if $x \geq 0$ and $\omega(x) = 0$ then $\rho_\omega^{1/2} x \rho_\omega^{1/2} \geq 0$ has trace 0, so $x = 0$); hence $q_a = 0$, i.e. $a^2 \in A$. Polarising, $a \circ b = \frac{1}{2}((a+b)^2 - a^2 - b^2) \in A$, so $P(a \circ b) = a \circ b$. $\square$

The faithfulness hypothesis is essential, not cosmetic: it is precisely what collapses the projected product onto the ambient one. Without it, an exact positive idempotent need not have a Jordan-subalgebra range, which is why Effros–Størmer must use $\bullet$ (Theorem 8.1).

Example 7.2 (Failure without faithfulness). (PROVED) On $\mathbb{R}^3$ with the sup norm let Φ act as the row-stochastic idempotent $P_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{3} & \frac{2}{3} & 0 \end{pmatrix}$ (equivalently $\Phi(X) = \text{diag}(P_0 \text{diag } X)$ on $L(\mathbb{C}^3)$). Then P_0 is a genuine unital positive idempotent. Its invariant states are $(t, 1-t, 0)$, $0 \leq t \leq 1$, so none is faithful (state 3 is transient). Its range is $\{v : v_3 = \frac{1}{3}(v_1 + 2v_2)\}$, and for $a = (-1, 1, \frac{1}{3}) \in \text{Im } P_0$ (with $\|a\|_\infty = 1$) one computes $a^2 = (1, 1, \frac{1}{9})$, $P_0(a^2) = (1, 1, 1)$, hence

$$q_a = P_0(a^2) - a^2 = (0, 0, \frac{8}{9}), \quad \|q_a\|_\infty = \frac{8}{9}.$$

The range is not closed under the ambient product; the hole is $\Theta(1)$.

The approximate transfer

The proof of Proposition 7.1 quantifies, reusing verbatim the square-hole estimate of the bridge (Lemma 6.5). The role of faithfulness is to convert “small expectation” into “small operator norm”, and this costs a factor $1/\lambda_\omega$.

Theorem 7.3 (Approximate ambient closure under a conditioned invariant state). (PROVED) *There are universal constants $\eta_0 > 0$, C such that the following holds. Let Φ be unital positive with $\|\Phi^2 - \Phi\| \leq \eta \leq \eta_0$, $P = \theta(2\Phi - \mathbf{1})$, $A = \text{Im } P$, and suppose Φ admits a faithful state ω with $\|\omega \circ \Phi - \omega\| \leq \eta$ and density $\rho_\omega \geq \lambda \mathbf{1}$, $\lambda > 0$. Then for all $a, b \in A$,*

$$\|a \circ b - P(a \circ b)\| = \|h_{a,b}\| \leq C \frac{\eta}{\lambda} \|a\| \|b\|. \quad (39)$$

In particular $\bullet$ and $\circ$ agree on A to within $O(\eta/\lambda)$.

Proof. Fix $a \in A$ and split $q_a = q^+ - q^-$ into orthogonal positive and negative parts. By (27) of Lemma 6.5, $q_a \geq -C\eta\|a\|^2 \mathbf{1}$, so $0 \leq q^- \leq C\eta\|a\|^2 \mathbf{1}$ and hence $\|q^-\| \leq C\eta\|a\|^2$. Since $\|P - \Phi\| \leq C\eta$ and $\|\omega \circ \Phi - \omega\| \leq \eta$,

$$\omega(q_a) = \omega(P(a^2)) - \omega(a^2) = \omega(a^2) - \omega(a^2) + O(\eta)\|a\|^2,$$

so $|\omega(q_a)| \leq C\eta\|a\|^2$, whence $\omega(q^+) = \omega(q_a) + \omega(q^-) \leq C\eta\|a\|^2$. Faithfulness upgrades this to an operator-norm bound: for $x \geq 0$, $\omega(x) = \text{Tr}(\rho_\omega x) \geq \lambda \text{Tr}(x) \geq \lambda\|x\|$, so applying this to $x = q^+$,

$$\|q^+\| \leq \frac{\omega(q^+)}{\lambda} \leq C \frac{\eta}{\lambda} \|a\|^2.$$

Therefore $\|q_a\| = \max(\|q^+\|, \|q^-\|) \leq C(\eta/\lambda)\|a\|^2$ (using $\lambda \leq 1$). Finally $(a, b) \mapsto h_{a,b}$ is a symmetric bilinear map with diagonal $h_{a,a} = -q_a$. For $t > 0$,

$$h_{a,b} = \frac{1}{4t}(h_{a+tb, a+tb} - h_{a-tb, a-tb}),$$

so the diagonal estimate gives

$$\|h_{a,b}\| \leq \frac{C\eta}{4t\lambda}(\|a+tb\|^2 + \|a-tb\|^2) \leq \frac{C\eta}{2t\lambda}(\|a\| + t\|b\|)^2.$$

Choosing $t = \|a\|/\|b\|$ when $a, b \neq 0$ (the zero cases being trivial) gives (39). $\square$

The hypothesis asks only that ω be *approximately* invariant ($\|\omega \circ \Phi - \omega\| \leq \eta$). The exact-invariance subcase $\omega \circ \Phi = \omega$ (with Φ still merely almost idempotent) yields the same bound (39) and was obtained independently by both authors; Theorem 7.3 is the approximate-invariance extension.

Bare existence is not enough

Theorem 7.3 is meaningful only when λ is bounded below *relative to* η : the bound is vacuous once $\lambda \lesssim \eta$. This is not an artefact of the proof. The colleague's question, taken literally (" Φ admits a faithful invariant state", with no quantitative modulus), has a negative answer, witnessed by a single explicit family in which every member is faithful yet the holes do not vanish.

Proposition 7.4 (Faithful invariant state alone does not transfer). (PROVED) *Let $J = \frac{1}{3}\mathbf{J}$ be the 3×3 all-ones-over-three matrix and P_0 as in Example 7.2. For $a \in (0, \frac{1}{2})$ set*

$$T_a = (1-a)P_0 + aJ, \quad \Phi_a(X) = \text{diag}(T_a \text{diag } X) \text{ on } L(\mathbb{C}^3).$$

Each T_a is strictly positive row-stochastic, hence Φ_a is unital positive with a faithful invariant state $\pi_a = (\frac{4}{9} - \frac{a}{9}, \frac{5}{9} - \frac{2a}{9}, \frac{a}{3})$, and

$$\eta_a = \|T_a^2 - T_a\| = \frac{10}{9}a(1-a) \xrightarrow{a \rightarrow 0} 0.$$

Writing P_a for the spectral idempotent onto the two eigenvalues 1 and $1-a$ near 1, one has $P_a r = r$ for $r = (1, 0, \frac{1}{3})$ but

$$\|(\mathbf{1} - P_a)(r^2)\|_\infty = \frac{2}{9} - \frac{2a}{27} \xrightarrow{a \rightarrow 0} \frac{2}{9}.$$

Thus the range carries an ambient-product hole bounded away from 0, despite exact faithful invariance and $\eta_a \rightarrow 0$. The conditioning degrades exactly in step: $\lambda = \pi_{a,3} = a/3 = \Theta(\eta_a)$, so $\eta_a/\lambda \rightarrow 10/3 = \Theta(1)$ and (39) is correctly vacuous.

The number $2/9$ and the projector identity $P_a = ((2-a)T_a - T_a^2)/(1-a)$ were verified numerically by both authors (matching the Riesz projector to machine precision). Proposition 7.4 settles the colleague's question: *the projected Effros–Størmer product $P(x \circ y)$ remains necessary; the ambient product cannot replace it under the bare hypothesis.*

What survives, and its scope

Remark 7.5 (Conditional upgrade, and why it is narrow). (PROVED, CONDITIONAL) If, in addition, the invariant state is *well-conditioned* with a dimension-free floor $\lambda = \Omega(1)$, then (39) reads $\|a \circ b - P(a \circ b)\| = O(\eta)\|a\|\|b\|$: the projected and ambient products agree to $O(\eta)$. Since the ambient

$\circ$ satisfies the Jordan identity *exactly*, substituting $\circ$ for $\bullet$ (finitely many $O(\eta)$ replacements with bounded factors) shows the Jordan-identity defect (20) is then $O(\eta)$ rather than $O(\sqrt{\eta})$ — the Kitaev-strength exponent of Remark 11.1, here obtained from a *variance* input (the faithful invariant state) rather than a dilation.

This is a genuine but *narrow* sufficient condition, not a general mechanism. The floor obeys $\lambda \leq 1/\dim \mathcal{H}$ (with equality for the maximally mixed state, giving only $O(\eta \dim \mathcal{H})$), and *nothing in the hypotheses bounds λ away from 0*: Proposition 7.4 exhibits a fixed-dimensional family, faithful throughout, with $\lambda \rightarrow 0$. Hence this does not advance the exponent question for general positive maps, which remains as in Remark 11.1. This quantitative floor should not be conflated with the qualitative faithful-state assumption in the van Luijk–Wilming fixed-point theory (Remark 12.1); their finite-dimensional faithfulness hypothesis supplies no dimension-free lower bound on λ .

8 Exact factorization by positive maps

The bridge of Section 6 turns an almost-idempotent positive map into an *approximate* Jordan algebra. The original goal of Kitaev’s theorem, however, is sharper: to *factor* the map, up to a small error, through a *genuine* algebra by structure-preserving maps. This section records what is known about the positive-map analogue of that factorization. The endpoint is classical (Effros–Størmer); the obstruction is that the spectral idempotent P is only approximately positive, and rounding it to a genuinely positive map cannot be done generically. The viable route is a stability statement for near-positive idempotents, conditional on which we obtain exact unital-positive factor maps with the bridge exponent (Theorem C below).

Throughout, $\Phi: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ is unital positive with $\|\Phi^2 - \Phi\| \leq \eta$, $P = \theta(2\Phi - \mathbf{1})$ is the spectral idempotent of Section 4, $A = \text{Im } P$, and $a \bullet b = P(a \circ b)$.

8.1 The exact template

The factorization we approximate is the exact theorem of Effros and Størmer: a positive unital projection makes its range a genuine Jordan algebra, and *is* the pair of factor maps.

Theorem 8.1 (Effros–Størmer). (CITED) *Let A be a unital JC-algebra and $E: A \rightarrow A$ a unital positive projection ($E^2 = E$). Then the range $E(A)$, equipped with the vector-space operations inherited from A and the product*

$$r \bullet s = E(r \circ s),$$

is a JC-algebra. Equivalently, the fixed-point set of any normal unital positive map of a JW-algebra into itself carries a natural JW-algebra structure.

Theorem 8.1 is the Jordan analogue of the Choi–Effros theorem [1] for completely positive projections, and the exact ($\eta = 0$) limit of the bridge (Remark 6.9). If the spectral idempotent P of an almost-idempotent Φ were *genuinely* positive, Theorem 8.1 would apply to $E = P$ directly and produce a genuine JC-algebra $J = P(B(\mathcal{H})_{\text{sa}})$ together with exact factor maps. The difficulty is that $P = \theta(2\Phi - \mathbf{1})$ is only $O(\eta)$ -positive ((22)), and may genuinely fail to be positive.

Remark 8.2 (P can fail to be positive). (PROVED) The approximate positivity in (22) cannot be upgraded to exact positivity: there is an explicit classical (commutative) family of positive unital almost-idempotent maps Φ on $\mathbb{R}^4$ for which $P = \theta(2\Phi - \mathbf{1})$ has strictly negative entries. Thus the projected product $a \bullet b = P(a \circ b)$ of the bridge is genuinely needed; one cannot simply invoke Theorem 8.1 for P .

8.2 Why generic positivity rounding fails

One might hope to repair P to a positive map abstractly: given a map that is ε -positive, find a genuinely positive map within $O(\varepsilon)$. *No such dimension-free repair exists*, even for the simplest non-commutative Jordan targets. The obstruction uses spin factors.

Definition 8.3 (Spin-factor cone). (CITED) For the spin factor $J = \mathbb{R}\mathbf{1} \oplus \mathbb{R}^m$ (Theorem 3.5), an element (s, v) with $s \in \mathbb{R}$, $v \in \mathbb{R}^m$ has square $(s, v) \circ (s, v) = (s^2 + \|v\|_2^2, 2sv)$ and spectrum $\{s \pm \|v\|_2\}$. Hence the positive cone and the order-unit norm are

$$J_+ = \{(s, v) : s \geq \|v\|_2\}, \quad \|(s, v)\| = |s| + \|v\|_2.$$

A linear map T into J is ε -positive if $T(x) \geq -\varepsilon\|x\|\mathbf{1}$ for $0 \leq x \leq \mathbf{1}$.

Proposition 8.4 (Failure of dimension-free positivity rounding). (PROVED; OBSTRUCTION) *For every integer $m \geq 1$ put $\varepsilon = 1/(2m)$ and let $J = \mathbb{R}\mathbf{1} \oplus \mathbb{R}^m$ be the spin factor. Define a unital map $T: \ell_{2m}^\infty \rightarrow J$ on the coordinate atoms $\delta_{j,\pm}$ ($j = 1, \dots, m$) by*

$$T(\delta_{j,+}) = (\varepsilon, 2\varepsilon e_j), \quad T(\delta_{j,-}) = (\varepsilon, -2\varepsilon e_j),$$

where $e_1, \dots, e_m$ is the standard basis of $\mathbb{R}^m$. Then T is unital and ε -positive, yet

$$\|T - S\| \geq \sqrt{\varepsilon} \quad \text{for every positive unital map } S: \ell_{2m}^\infty \rightarrow J.$$

The same lower bound holds with domain $B(\mathbb{C}^{2m})_{\text{sa}}$, by composing with the diagonal conditional expectation.

Proof idea. Unitality of T holds because the $2m$ atom-images sum to $(1, 0) = \mathbf{1}$; ε -positivity follows from $2\|v\|_2 \leq \sum_j p_j + 1$ for a sub-probability vector p . For the lower bound: a *positive* unital S has atom-images $S(\delta_i) = (s_i, w_i) \in J_+$ with $s_i \geq \|w_i\|_2$ and $\sum_i (s_i, w_i) = (1, 0)$, whence $\sum_i \|w_i\|_2 \leq \sum_i s_i = 1$. The vector parts of T have $\sum_i \|v_i\|_2 = 2$. Averaging over Rademacher signs $\sigma_i \in \{\pm 1\}$, $\mathbb{E}_\sigma \|\sum_i \sigma_i (v_i - w_i)\|_2^2 = \sum_i \|v_i - w_i\|_2^2 \geq \varepsilon$, so some sign pattern gives $\|\sum_i \sigma_i (v_i - w_i)\|_2 \geq \sqrt{\varepsilon}$, which is dominated by $\|T - S\|$ tested against the corresponding element of ℓ_{2m}^∞ . $\square$

Proposition 8.4 shows that an approximate *algebra* isomorphism alone cannot deliver exact positive factor maps with the bridge exponent: a black-box $O(\varepsilon)$ repair is impossible, and any generic repair must lose at least a square root. Note the obstruction needs a *free* map (no retraction constraint): the construction does not respect an idempotent relation $S \circ \iota = \text{id}$. This is exactly the loophole the next subsection exploits. In the fully commutative case (both domain and codomain commutative) the obstruction is absent: normalising the positive parts of signed rows yields a positive unital map within 2ε (this is the starting point of Section 9).

8.3 Near-positive projection stability

The escape from Proposition 8.4 is to ask for stability of *idempotents*, not of arbitrary near-positive maps. An idempotent's retraction constraint pins it to its own range and rules out the cheap sign-spreading used in the counterexample.

Open Problem 8.5 (Near-positive projection stability). (OPEN; SHARP EXPONENT 1/2) *There exist universal constants $\delta_0, K > 0$ (independent of dimension) such that the following holds for every finite-dimensional $\mathcal{H}$. If $R: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ is linear with*

$$R^2 = R, \quad R(\mathbf{1}) = \mathbf{1}, \quad \|R\| \leq 1 + \delta, \quad R(x) \geq -\delta \mathbf{1} \quad \text{for } 0 \leq x \leq \mathbf{1},$$

and $0 < \delta \leq \delta_0$, then there is a unital positive idempotent $E: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ with

$$\|E - R\| \leq K\sqrt{\delta}.$$

The square-root exponent is the best possible: linear ($O(\delta)$) stability is already false in the commutative case (Section 9, Hume's example Example 9.4). Open Problem 8.5 is open in general; its commutative reduction and the special cases now proved are the subject of Section 9.

8.4 Conditional exact factorization (Theorem C)

Granting Open Problem 8.5, the factorization goes through with the bridge exponent and *genuinely positive* factor maps.

Theorem 8.6 (Conditional exact unital-positive factorization). (PROVED, CONDITIONAL ON OPEN PROBLEM 8.5) *Assume Open Problem 8.5. Then there are universal constants $\eta_0, C > 0$ such that the following holds. Let $\mathcal{H}$ be finite-dimensional and $\Phi: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ unital positive with $\|\Phi^2 - \Phi\| \leq \eta \leq \eta_0$. Then there exist a finite-dimensional special JB-algebra J (a JC-algebra) and unital positive maps*

$$\Delta: J \rightarrow B(\mathcal{H})_{\text{sa}}, \quad \Upsilon: B(\mathcal{H})_{\text{sa}} \rightarrow J,$$

such that

$$\Upsilon\Delta = \text{id}_J, \quad \|\Delta\Upsilon - \Phi\| \leq C\sqrt{\eta}, \quad \Upsilon(\Delta(x) \circ \Delta(y)) = x \bullet y \quad (x, y \in J),$$

where $\bullet$ is the Jordan product of J . Concretely Δ is the inclusion of J into $B(\mathcal{H})_{\text{sa}}$ and $\Upsilon = E$ is a unital positive idempotent with $J = E(B(\mathcal{H})_{\text{sa}})$ and $x \bullet y = E(x \circ y)$.

Proof. Form $P = \theta(2\Phi - \mathbf{1})$. By the spectral estimate of Section 4 (see (21)–(22)), P is a unital idempotent with $\|P - \Phi\| \leq C\eta$, $\|P\| \leq 1 + C\eta$, and $P(x) \geq -C\eta \mathbf{1}$ for $0 \leq x \leq \mathbf{1}$. Thus P satisfies the hypotheses of Open Problem 8.5 with $\delta = C\eta$ when $\eta > 0$. If $\eta = 0$, then $P = \Phi$ is already a unital positive idempotent and we set $E = P$. Otherwise apply Open Problem 8.5: there is a unital positive idempotent E with $\|E - P\| \leq K\sqrt{C\eta} \leq C'\sqrt{\eta}$.

Extend E complex-linearly to $E_{\mathbb{C}}(a + ib) = E(a) + iE(b)$ on $B(\mathcal{H})$; since positivity is tested on positive self-adjoint inputs, $E_{\mathbb{C}}$ is a unital positive projection on the JC-algebra $B(\mathcal{H})$. By Theorem 8.1, $J := E(B(\mathcal{H})_{\text{sa}})$ is a JC-algebra under $x \bullet y = E(x \circ y)$; it is special by construction as the Effros–Størmer JC range inside $B(\mathcal{H})$ with projected product $E(x \circ y)$, so the exceptional Albert algebra cannot occur. This does not assert closure under the ambient product $\circ$.

Set $\Delta =$ the inclusion $J \hookrightarrow B(\mathcal{H})_{\text{sa}}$ and $\Upsilon = E$. Both are unital and positive: Δ maps the cone $J_+ = J \cap B(\mathcal{H})_+$ into $B(\mathcal{H})_+$, and $E(B(\mathcal{H})_+) \subseteq J_+$ because E is positive and idempotent. Then $\Upsilon\Delta = E|_J = \text{id}_J$, and

$$\|\Delta\Upsilon - \Phi\| = \|E - \Phi\| \leq \|E - P\| + \|P - \Phi\| \leq C'\sqrt{\eta} + C\eta \leq C\sqrt{\eta}.$$

The multiplicativity identity $\Upsilon(\Delta x \circ \Delta y) = E(x \circ y) = x \bullet y$ is the definition of the Effros–Størmer product. Finally the JB positive cone of $(J, \bullet)$ equals the inherited cone $J \cap B(\mathcal{H})_+$: if $x = y \bullet y = E(y \circ y)$ then $x \geq 0$ ambiently by positivity of E ; conversely $z \in J \cap B(\mathcal{H})_+$ with $m = \|z\|$ satisfies $\|m\mathbf{1} - z\| \leq m$, which by the unital JB norm criterion makes z a positive element of $(J, \bullet)$. Hence the factor maps are positive for the intrinsic JB order of J . $\square$

Theorem 8.6 isolates the entire remaining difficulty in a single unproved input, Open Problem 8.5: *the implication is proved*; only the stability theorem is open. There is no use of abstract positivity

rounding (which Proposition 8.4 forbids) and no use of the abstract structure theorem of Section 10; the route is independent of Layer 1. The price is the exponent: $\sqrt{\eta}$ rather than η , which by Hume's example (Example 9.4) is the best this route can give.

The factor maps in Theorem 8.6 are unital positive maps. The theorem does *not* assert decomposable factor maps in general. Such a strengthening requires additional reversibility or universal-reversibility input on the exact JC range, as in the exact theory discussed around Section 11.

Remark 8.7 (An alternative route through Layer 1). (OPEN) If instead the abstract structure theorem (Section 10) were proved with a *positivity-capable* output — a genuine JC-subalgebra $J_0 \subseteq B(\mathcal{H})_{\text{sa}}$ whose points are close to the bridge range $A = \text{Im } P$, and a canonical positive conditional expectation E_{J_0} close to P — then comparing P to E_{J_0} would give exact positive factor maps without invoking Open Problem 8.5. This route is not yet formalised; algebraic closeness alone is insufficient, precisely because of Proposition 8.4.

9 The commutative case: stability of almost-idempotent stochastic maps

The exact factorization of Section 8 rests on one open input, near-positive projection stability (Open Problem 8.5). This section develops that problem in the *commutative* (classical) setting, where it becomes a concrete and appealing question about stochastic matrices, and records the substantial partial progress: the sharp exponent, several proved special cases, an exact commutative factorization theorem under a checkable geometric hypothesis, and the precise remaining obstruction. All constants in this section are universal (independent of the matrix size n) unless stated otherwise.

9.1 The classical reduction

A unital positive map of the commutative algebra $\ell_n^\infty = \mathbb{R}^n$ into itself is a row-stochastic matrix; the bridge data specialise as follows.

Definition 9.1 (Stochastic maps and retractions). (CONSENSUS) A real $n \times n$ matrix Q is *row-stochastic* if $Q \geq 0$ entrywise and $Q\mathbf{1} = \mathbf{1}$ (rows sum to 1), where $\mathbf{1} = (1, \dots, 1)$. Equivalently Q is the matrix of an affine self-map of the probability simplex Δ_n . Its non-idempotence is measured by $\|Q^2 - Q\|_{\infty \rightarrow \infty}$, the maximum row ℓ^1 -norm of $Q^2 - Q$. A *stochastic idempotent* is a row-stochastic E with $E^2 = E$; these are exactly the affine retractions of Δ_n onto a sub-polytope. A *signed affine retraction* is a matrix P with $P\mathbf{1} = \mathbf{1}$ and $P^2 = P$ (rows are signed measures of total mass 1); its *negative mass* is $\text{neg}(p_i) = \sum_j \max(-p_{ij}, 0)$, and we write $\delta = \max_i \text{neg}(p_i)$. The *row polytope* is $K = \text{conv}\{p_1, \dots, p_n\}$, the convex hull of the rows.

Open Problem 9.2 (Classical projection stability). (OPEN; SHARP EXPONENT 1/2) *There are universal constants $\eta_0, C > 0$ such that every row-stochastic Q with $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta \leq \eta_0$ admits a stochastic idempotent E with $\|Q - E\|_{\infty \rightarrow \infty} \leq C\sqrt{\eta}$.*

Open Problem 9.2 is the commutative case of Open Problem 8.5. Within the commutative setting, the stochastic formulation and the signed-retraction formulation are equivalent up to universal constants:

Lemma 9.3 (Equivalence of the two formulations). (PROVED) (i) *If Q is row-stochastic with $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta$ then $P = \theta(2Q - \mathbf{1})$ is a signed affine retraction with $\|P - Q\|_{\infty \rightarrow \infty} \leq C\eta$ and negative*

mass $\delta \leq C\eta$. (ii) Conversely, if P is a signed affine retraction with negative mass δ , then the row-normalised positive parts $q_i = p_i^+ / \|p_i^+\|_1$ form a row-stochastic Q with

$$\|P - Q\|_{\infty \rightarrow \infty} \leq 2\delta, \quad \|Q^2 - Q\|_{\infty \rightarrow \infty} \leq 6\delta + 4\delta^2.$$

Hence Open Problem 9.2 for Q and the signed-retraction stability “ $\text{dist}(P, \text{stochastic idempotents}) \leq C\sqrt{\delta}$ ” imply one another with universal constants.

The convenience of the signed form is that $P^2 = P$ holds *exactly*; idempotency can be used as an algebraic identity, with the cost of non-positivity quarantined in the small parameter δ . The geometric content of $P^2 = P$ is that the rows are fixed by P acting on the right, and the affine functions on the row polytope interact rigidly with this fixity — the mechanism behind every estimate below.

9.2 The exponent is sharp

Example 9.4 (Hume’s sharp family). (PROVED; SHARPNESS) For small $s > 0$ set $v_s = (1, -1 + s, -s)$, $u_s = (1 - s + s^2, -s, 0)^\top$, and $P_s = I - u_s v_s^\top$. Then $P_s \mathbf{1} = \mathbf{1}$, $P_s^2 = P_s$, the negative mass is $\delta = s^2$, and the distance from P_s to *every* stochastic idempotent is

$$2s - 2s^2 + 2s^3 = 2\sqrt{\delta} + O(\delta).$$

Hence no bound $\text{dist}(P, \cdot) \leq C\delta^\beta$ with $\beta > \frac{1}{2}$ can hold: the exponent $\frac{1}{2}$ in Open Problems 8.5 and 9.2 is optimal.

9.3 The two core lemmas

Both lemmas express the rigidity of $P^2 = P$ (equivalently $Q^2 \approx Q$) against affine test functions. We write $\tau = \sqrt{\delta}$ (or $\sqrt{\eta}$). An affine function $h: K \rightarrow [0, 1]$ is called 1-*Lipschitz* for the row ℓ^1 -geometry if $|h(x) - h(y)| \leq \|x - y\|_1$.

Lemma 9.5 (Affine-face leakage). (PROVED) Let Q be row-stochastic with $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta$, let $h: \Delta_n \rightarrow [0, 1]$ be affine, put $m = \max_j h(q_j)$ and $d_i = m - h(q_i) \geq 0$. Then for every row i and every $\gamma > 0$,

$$q_i(\{j : h(q_j) \leq m - \gamma\}) \leq \frac{d_i + \eta}{\gamma}.$$

In particular a maximiser row ($d_i = 0$) sends at most $\sqrt{\eta}$ of its mass below the level $m - \sqrt{\eta}$: exposed faces are almost closed at the square-root scale, and no linear ($O(\eta)$) closure is possible.

Proof idea. Right-fixity $q_i Q = q_i$ (a row of $Q^2 \approx Q$) gives $\sum_j q_i(j) h(q_j) = h(q_i) + O(\eta)$. Since $0 \leq h \leq 1$ and $h(q_j) \leq m$, the mass on $\{h(q_j) \leq m - \gamma\}$ is penalised by at least γ per unit, forcing it below $(d_i + \eta)/\gamma$. Taking $\gamma = \sqrt{\eta}$ gives the sharp scale. $\square$

To localise a row at a *vertex* (not merely a face) one needs the vertex to be exposed, with a quantitative gap.

Definition 9.6 (Exposed vertex and exposedness modulus). (CONSENSUS) A row vertex v of K is (ρ, κ) -*exposed* if there is an affine $h: K \rightarrow [0, 1]$ with $h(v) = 0$ and $h(p_i) \geq \kappa$ for every row p_i with $\|p_i - v\|_1 \geq \rho$. Thus exposedness is measured on the outside *row set*, not on every point of K . The *exposedness modulus* is

$$e_v(\rho) = \sup_h \min_{\substack{i \\ \|p_i - v\|_1 \geq \rho}} h(p_i),$$

the supremum over affine $h: K \rightarrow [0, 1]$ with $h(v) = 0$ (with the empty minimum interpreted as 1). The vertex is *well-exposed at the square-root scale* if $e_v(\rho) \geq c\sqrt{\delta}$ for some $\rho = O(\sqrt{\delta})$.

Lemma 9.7 (Exposed-vertex concentration and circuit cancellation). (PROVED) *Let P be a signed affine retraction with negative mass $\leq \delta$. (i) (Concentration.) If a row vertex v is (ρ, κ) -exposed, then its positive part concentrates near v : writing $U_v = \{j : \|p_j - v\|_1 < \rho\}$, there is a probability vector π_v supported on U_v with*

$$\|v - \pi_v\|_1 \leq C\left(\frac{\delta}{\kappa} + \delta\right).$$

(ii) (Circuit cancellation.) If $v_1, \dots, v_k$ are pairwise $\|\cdot\|_1$ -separated row vertices, each (ρ, κ) -exposed, then for all real coefficients c_a ,

$$\left\| \sum_{a=1}^k c_a v_a \right\|_1 \geq \left(1 - C\left(\frac{\delta}{\kappa} + \delta\right)\right) \sum_{a=1}^k |c_a|.$$

For $\kappa \geq c\sqrt{\delta}$ both right-hand sides are $1 - O(\sqrt{\delta})$.

Proof idea. (i) Applying the exposing h to the right-fixity identity $\sum_j v_j h(p_j) = 0$ and splitting $v = v^+ - v^-$ forces $v^+(U_v^c) \leq \delta/\kappa$ (mass far from v is penalised by κ); truncating and normalising the positive part on U_v gives π_v . (ii) The U_a are pairwise disjoint by separation, so the π_a have disjoint supports and $\|\sum c_a \pi_a\|_1 = \sum |c_a|$; the triangle inequality with (i) transfers this to the v_a . $\square$

9.4 Proved special cases

The following classes of P (equivalently Q) satisfy Open Problem 9.2 with universal constants in their stated regimes. They are not a single nested chain: the rank-one and line-segment results are complementary, line-segments are the one-dimensional simplex case, well-exposed separated vertices force a simplex but are not equivalent to arbitrary simplex geometry, and the cluster theorem packages exposed representative geometries with transient rows.

Theorem 9.8 (Rank-one stability). (PROVED) *There are universal constants $\delta_0, C > 0$ such that, if $0 \leq \delta \leq \delta_0$ and $P = I - uv^\top$ is a signed affine retraction (so $\sum_j v_j = 0$, $v^\top u = 1$) with negative mass $\leq \delta$, there is a stochastic idempotent E with $\|P - E\|_{\infty \rightarrow \infty} \leq C\sqrt{\delta}$. This class contains Hume's sharp family (Example 9.4).*

Proof idea. Normalising the positive and negative masses of v , the negative-mass bound forces at most one positive and one negative coordinate of u to exceed $\sqrt{\delta}$ in size; the corresponding two-state rounding gives a stochastic idempotent within $C\sqrt{\delta}$. The $\sqrt{\delta}$ arises because controlling a product $u_i v_j$ by δ forces one factor to be $O(\sqrt{\delta})$. $\square$

Theorem 9.9 (Line-segment and simplex stability). (PROVED) *There are universal constants $\delta_0, C > 0$ such that the following holds for every signed affine retraction P with negative mass $\leq \delta \leq \delta_0$. (i) If the row polytope K has affine dimension ≤ 1 (a point or segment), there is a stochastic idempotent E with $\|P - E\|_{\infty \rightarrow \infty} \leq C\sqrt{\delta}$. (ii) If K is a simplex with vertices $r^1, \dots, r^m$ among the rows, the same conclusion holds, with a constant C **independent of the number of vertices m and of n** .*

Proof idea. For a simplex, each row has unique barycentric coordinates $\lambda_a(\cdot)$. Right-fixity $r^a P = r^a$ tested against λ_b gives $\sum_j r_j^a \lambda_b(j) = \delta_{ab}$, hence $\sum_j r_j^a (1 - \lambda_a(j)) = 0$; near-positivity then concentrates the positive mass of r^a on its near-vertex slice $U_a = \{\lambda_a \geq 1 - \sqrt{\delta}\}$, and these slices are pairwise disjoint. Normalising the slice restrictions to probability vectors π_a and rounding every row to the same barycentric mixture of the π_a produces a stochastic idempotent within $C\sqrt{\delta}$. No vertex-count factor appears because the barycentric decomposition is convex. $\square$

Theorem 9.10 (Well-exposed vertices force a simplex). (PROVED) *There are universal constants $\delta_0, c, C > 0$ such that the following holds. Let P be a signed affine retraction with negative mass $\leq \delta \leq \delta_0$. If every vertex of the row polytope K is pairwise separated and (ρ, κ) -exposed with $\rho \leq C\sqrt{\delta}$ and $\kappa \geq c\sqrt{\delta}$, then the vertices are affinely independent — so K is a simplex — and Theorem 9.9 gives a stochastic idempotent within $C\sqrt{\delta}$.*

Proof idea. A nontrivial affine dependence $\sum_a c_a v_a = 0$ (with $\sum c_a = 0$, $\sum |c_a| > 0$) would contradict the circuit-cancellation bound of Lemma 9.7(ii), which forces $\|\sum c_a v_a\|_1 > 0$. $\square$

Theorem 9.11 (Cluster-representative stability). (PROVED) *There are universal constants $\delta_0, C > 0$ such that the following holds for every signed affine retraction P with negative mass $\leq \delta \leq \delta_0$. Suppose there are representative rows $r^1, \dots, r^m$ with parameters $\rho, \kappa, \gamma > 0$ such that: the r^a are pairwise 2ρ -separated; each r^a is (ρ, κ) -exposed (so the clusters $U_a = \{j : \|p_j - r^a\|_1 < \rho\}$ are disjoint); and every row outside all clusters is within γ of $\text{conv}\{r^a\}$. Then there is a stochastic idempotent E with*

$$\|P - E\|_{\infty \rightarrow \infty} \leq C\left(\rho + \gamma + \frac{\delta}{\kappa} + \delta\right).$$

In particular $\rho, \gamma = O(\sqrt{\delta})$ and $\kappa \geq c\sqrt{\delta}$ give the $O(\sqrt{\delta})$ bound, with C independent of m, n , and the number of transient rows.

Proof idea. Concentrate each representative to π_a on U_a via Lemma 9.7(i); build E with rows π_a on the clusters and the prescribed convex combination of the π_a off the clusters. Disjoint supports give $E^2 = E$; the error is the sum of the cluster radius, the reconstruction error, and the concentration error, with no accumulation over m because the reconstruction coefficients form probability vectors. $\square$

9.5 Exact commutative factorization

The cluster-representative theorem packages directly into an exact positive-map factorization — the commutative instance of Theorem 8.6, but now *unconditional* apart from a checkable geometric hypothesis on the spectral idempotent.

Theorem 9.12 (Exact commutative factorization). (PROVED, UNDER THE CLUSTER GEOMETRY HYPOTHESIS) *There are universal constants $\eta_0, C > 0$ such that the following holds. Let Q be row-stochastic with $\eta = \|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta_0$, and suppose the rows of $P = \theta(2Q - \mathbf{1})$ satisfy the cluster-representative geometry of Theorem 9.11 with $\rho, \gamma = O(\sqrt{\eta})$, $\kappa \geq c\sqrt{\eta}$. Then there exist a finite-dimensional commutative special JB-algebra J and unital positive maps $\Delta: J \rightarrow \ell_n^\infty$, $\Upsilon: \ell_n^\infty \rightarrow J$ with*

$$\Upsilon \Delta = \text{id}_J, \quad \|\Delta \Upsilon - Q\|_{\infty \rightarrow \infty} \leq C\sqrt{\eta}, \quad \Upsilon(\Delta x \cdot \Delta y) = x \bullet y,$$

where $\cdot$ is pointwise product and $\bullet$ the JB product of J .

Proof idea. Theorem 9.11 gives a stochastic idempotent E with $\|E - Q\| \leq C\sqrt{\eta}$. Set $J = E(\ell_n^\infty)$; by Theorem 8.1 (commutative case) J is a JB-algebra under $x \bullet y = E(xy)$. Take $\Delta = \text{inclusion}$, $\Upsilon = E$. The same hypothesis admits a *global exposed-hull* form: if every row of P is within $O(\sqrt{\eta})$ of $\text{conv } W_{\rho, \kappa}$, where $W_{\rho, \kappa}$ is the set of (ρ, κ) -exposed vertices, the conclusion holds (pass to a maximal 4ρ -separated subset and apply Theorem 9.11). $\square$

9.6 Reductions toward the general case

Two reductions show how close the general theorem is, and where it is stuck.

Proposition 9.13 (Approximate-simplicity reduction). (PROVED; REDUCES OPEN PROBLEM 9.2 TO A COORDINATE HYPOTHESIS) *Suppose the rows of P admit γ -approximate simplex coordinates: representatives r^a and affine λ_a with $\sum_a \lambda_a = 1$, $\lambda_a(r^b) = \delta_{ab}$ and $\|p_i - \sum_a \lambda_a(p_i)r^a\|_1 \leq \gamma$. If the λ_a are nonnegative (or, more generally, have coefficient negative mass $O(\delta)$), then there is a stochastic idempotent E with $\|P - E\|_{\infty \rightarrow \infty} \leq C(\sqrt{\delta} + \gamma)$. Consequently Open Problem 9.2 follows from proving that every near-positive signed affine retraction has $\gamma = O(\sqrt{\delta})$ approximate simplex coordinates with $O(\delta)$ coordinate negative mass.*

The robustness ($O(\delta)$ coordinate negative mass suffices, not exact nonnegativity) is essential: it converts the target from a positivity statement into a metric one. The reduction is sharp in the sense that coordinate negative mass $O(\sqrt{\delta})$ would only give $O(\delta^{1/4})$, so the $O(\delta)$ coordinate bound is the genuine target.

A second route rules out non-simplex geometry whenever the offending affine dependence has bounded affine witnesses.

Proposition 9.14 (Bounded-coordinate dependencies are impossible). (PROVED) *Suppose rows $r_1, \dots, r_k$ have distinct binary codewords realised by affine functions $s_1, \dots, s_m: K \rightarrow [0, 1]$ (i.e. $s_\ell(r_a) \in \{0, 1\}$). Then for all coefficients c_a ,*

$$\left\| \sum_a c_a r_a \right\|_1 \geq (1 - Cm\sqrt{\delta}) \sum_a |c_a|.$$

In particular, for $m\sqrt{\delta}$ small, the rows are affinely independent: an exact parallelogram row polytope ($r_{00} + r_{11} = r_{10} + r_{01}$) and, more generally, any bounded-complexity product-of-simplices geometry with affine $[0, 1]$ -coordinates, cannot occur. The constant degrades with the number m of coordinate bits, so this does not by itself settle geometries needing unboundedly many witnesses.

9.7 The remaining obstruction

What is left open is exactly the gap between “well-exposed/simplex” (proved) and arbitrary row polytopes. The cleanest sufficient statement is:

Open Problem 9.15 (Global exposed-hull lemma). (OPEN) *There are universal $c, C > 0$ such that for every near-positive signed affine retraction with negative mass $\leq \delta$, taking $\rho = C\sqrt{\delta}$ and $W_{\rho, \kappa} = \{\text{vertices } v : e_v(\rho) \geq \kappa\}$ with $\kappa = c\sqrt{\delta}$, every row is within $C\sqrt{\delta}$ of $\text{conv } W_{\rho, \kappa}$.*

By Theorem 9.12 (global form), Open Problem 9.15 implies Open Problem 9.2, hence the exact commutative factorization unconditionally. The difficulty is genuinely global: a *pointwise* exposed-or-redundant dichotomy (each vertex is either well-exposed or $O(\sqrt{\delta})$ -redundant) is *not* sufficient, because redundancy can be circular and sequential deletion can accumulate error.

Remark 9.16 (The regular-polygon warning, and why it is not a counterexample). (PROVED) A dense regular m -gon, as an abstract convex body, makes pointwise deletion accumulate: every vertex is $O(\sqrt{\delta})$ -redundant yet none is well-exposed at the $\sqrt{\delta}$ scale, so $W_{\rho,\kappa}$ can be empty. This shows the pointwise dichotomy is insufficient and forces the global formulation of Open Problem 9.15. However, regular polygons *cannot* arise as the row polytope of a near-positive signed affine retraction. Symmetrising any affine coordinate system over the dihedral group does not increase its maximum negative mass, and every resulting dihedrally-symmetric coordinate system on a fixed-diameter regular m -gon has vertex negative mass at least

$$\frac{\sqrt{3}}{\pi} - \frac{1}{3} \approx 0.218 - o_m(1),$$

bounded away from 0. So the regular polygon is a logical warning about the quantifier structure, not a genuine obstruction; a true counterexample to Open Problem 9.15 would need non-regular geometry with its offending feature shrunk to the $O(\sqrt{\delta})$ scale, at which point it no longer violates the desired conclusion.

9.8 Status of the commutative program

Table 1 summarises. Extensive low-dimensional searches and probes (random 3×3 families, four-row quadrilaterals, tensor powers of Hume’s family, direct sums, and non-simplex probes) have found *no* counterexample to Open Problem 9.2; the obstruction is the dimension-free reconstruction of an exposed skeleton, not a suspected failure of the conjecture.

Statement	Status
Equivalence of signed/stochastic forms (Lemma 9.3)	proved
$\sqrt{\cdot}$ exponent is sharp (Hume, Example 9.4)	proved
Affine-face leakage; exposed-circuit cancellation	proved
Rank-one; line-segment; simplex (vertex-count-free)	proved
Well-exposed $\Rightarrow$ simplex; cluster-representative	proved
Exact commutative factorization under cluster geometry (Theorem 9.12)	proved (conditional on geometry)
Approximate-simplicity / robust reduction (Proposition 9.13)	proved (reduction)
Bounded-coordinate dependencies impossible (Proposition 9.14)	proved
Global exposed-hull lemma (Open Problem 9.15)	open
Full classical projection stability (Open Problem 9.2)	open

Table 1: Status of the commutative projection-stability program.

10 The abstract structure theorem (Layer 1)

The bridge (Section 6) produces an ε -JB algebra; the factorization routes (Sections 8 and 9) bypass the abstract structure theory for the channel application. This section concerns the standalone

abstract structure theorem — Kitaev’s main theorem, Jordan-ised — which says that an ε -JB algebra is, up to a controlled deformation, a genuine JB-algebra. It is open. We state the target, the cohomological strategy, the one quantitative obstruction, and the substantial recent progress: a dimension-free *exact-adjoint splitting benchmark* that resolves the obstruction at the level of exact cocycles for every simple factor, together with a precise account of what still separates the benchmark from the full theorem.

10.1 The target

Open Problem 10.1 (Jordan structure theorem). (OPEN) *There are universal constants $C, \varepsilon_0 > 0$ (independent of dimension) such that for every $\varepsilon < \varepsilon_0$ and every finite-dimensional ε -JB algebra $\mathcal{A}$ there exist a genuine finite-dimensional JB-algebra $\mathcal{B}$ and a linear bijection $v: \mathcal{B} \rightarrow \mathcal{A}$ that is a $C\varepsilon$ -Jordan-isomorphism:*

$$\|v(\mathbf{1}_{\mathcal{B}}) - \mathbf{1}_{\mathcal{A}}\| \leq C\varepsilon, \quad \|v(b \bullet b') - v(b) \bullet v(b')\| \leq C\varepsilon \|b\| \|b'\|,$$

and $(1 - C\varepsilon)\|b\| \leq \|v(b)\| \leq (1 + C\varepsilon)\|b\|$ for all $b, b' \in \mathcal{B}$.

This is Kitaev’s statement [6, Theorem `th_main`] with the associative product replaced by the Jordan product and “ C^* -algebra” by “JB-algebra”. What we can state is the strategy and the precise point where the dimension-free constant is in question.

10.2 Strategy and vocabulary

Kitaev builds the model algebra $\mathcal{B}$ incrementally and controls the error at each step cohomologically. The Jordan programme mirrors four parts. An *idempotent element* is e with $e \bullet e = e$; a *Jordan frame* is a list of nonzero pairwise-orthogonal idempotents summing to $\mathbf{1}$ (the analogue of diagonal matrix units); the *Peirce decomposition* of an idempotent e splits $\mathcal{A}$ into the $1, \frac{1}{2}, 0$ eigenspaces of $L_e: a \mapsto e \bullet a$; a *derivation* is a linear D with $D(a \bullet b) = Da \bullet b + a \bullet Db$ (the D form the Lie algebra of $\text{Aut}(\mathcal{A})$).

- (a) **Approximate idempotents / Jordan frame.** Produce elements with $\|e \bullet e - e\|$ small, not near 0 or $\mathbf{1}$. The spectral idempotent P of Section 4 is a *map-level* device; it does not by itself produce idempotent *elements* of A , so this is a separate open step (in the channel application one instead uses the ambient $B(\mathcal{H})_{\text{sa}}$).
- (b) **Approximate Peirce decomposition.** For an approximate idempotent L_e has spectrum clustered near $\{0, \frac{1}{2}, 1\}$, giving an approximate Peirce splitting with $O(\varepsilon)$ cross-terms.
- (c) **Approximate coordinatization.** On a frame of rank ≥ 3 the Jordan–von Neumann–Wigner coordinatization [4] identifies the algebra with Hermitian matrices over a coordinate $*$ -algebra; the approximate version reads off coordinates up to $O(\varepsilon)$, producing a candidate $\mathcal{B}$ and a linear inclusion $v: \mathcal{B} \rightarrow \mathcal{A}$ multiplicative up to a (possibly large) δ .
- (d) **Error reduction.** Replace v by a nearby map whose defect is $O(\varepsilon)$, *uniformly in dimension*. This is the cohomological heart, treated below.

10.3 The cohomological error reduction

Write $g(b, b') = v(b \bullet b') - v(b) \bullet v(b')$ for the multiplicativity defect of the inclusion v from step (c); it is symmetric with $\|g\| \leq \delta$. Linearizing the Jordan identity ($\mathcal{B}$ exact, $\mathcal{A}$ only ε -JB) shows g is an

ε -approximate Jordan 2-cocycle. The Jordan 2-coboundary of a linear $h: \mathcal{B} \rightarrow \mathcal{A}$ is

$$(d^1 h)(b, b') = v(b) \bullet h(b') + h(b) \bullet v(b') - h(b \bullet b').$$

All cochain norms below are injective order-unit norms:

$$\|\theta\|_{\text{inj}} = \sup_{\|x_i\| \leq 1} \|\theta(x_1, \dots, x_k)\|.$$

We write $\text{dist}(\theta, \text{Im } d^1)$ for the distance in this norm. For an exact product $\circ$ and a unit-normalized symmetric 2-cochain $\theta(\mathbf{1}, x) = 0$, the linearized Jordan-identity defect is

$$\begin{aligned} (J\theta)(a, b) &= \theta(a^2, a \circ b) + a^2 \circ \theta(a, b) + (a \circ b) \circ \theta(a, a) \\ &\quad - \theta(a^2 \circ b, a) - a \circ \theta(a^2, b) - a \circ (b \circ \theta(a, a)). \end{aligned}$$

This is the first-order part of $(a * a) * (b * a) - ((a * a) * b) * a$ for $x * y = x \circ y + \theta(x, y)$; the informal symbol $d^2\theta$ in older internal notes is not used as a separate cohomological differential here. Below the actual first-order operator is always $J\theta$ (with quadratic error terms handled separately in the Newton step). Two ingredients drive a Newton iteration:

- a *splitting* (right inverse) s of d^1 with $\|s\| \leq K$: a recipe $h = sg$ with $d^1 h = g$ on exact cocycles;
- a *next-arrow (cocycle-projection) estimate*: every approximate cocycle θ is within $K'\|J\theta\|$ of an exact one, so that g may be corrected.

Setting $\tilde{v} = v - h$ replaces δ by $O(\delta^2 + (K + K')\varepsilon)$; iterating the Newton step

$$\delta \longmapsto C(K, K')(\varepsilon + \delta^2)$$

drives small δ to $O(\varepsilon)$. *The theorem reduces to: are K, K' dimension-free, in the order-unit norm?* (The Newton bookkeeping itself is verified, conditional on K, K' .)

The two qualitative algebraic inputs are classical.

Theorem 10.2 (Whitehead lemmas for Jordan algebras). (CITED; SOURCE-EXTRACTED, PRIMARY BYTE-CHECK PENDING) *Let J be a finite-dimensional semisimple Jordan algebra over a field of characteristic 0 and M a finite-dimensional J -module. Then every derivation $J \rightarrow M$ is inner ($H^1(J, M) = 0$, Jacobson); and if J is separable, every Jordan 2-cocycle is a coboundary ($H^2(J, M) = 0$, Penico).*

Proposition 10.3 (Compactness of $\text{Aut}(J)$). (CITED; SOURCE-EXTRACTED, PRIMARY BYTE-CHECK PENDING) *For a finite-dimensional Euclidean Jordan algebra J , $\text{Aut}(J)$ is a compact subgroup of the orthogonal group of the trace inner product; every automorphism preserves spectra, hence the order-unit norm. Normalized Haar averaging over $\text{Aut}(J)$ is therefore available.*

The vanishing $H^1 = H^2 = 0$ makes the correction step possible *in principle*; the issue is its *size*. Crucially, Haar averaging over $\text{Aut}(J)$ is a norm-one *projection* onto invariants but is *not* a right inverse to d^1 ; the inverse on the non-invariant cochain components is exactly where dimension can enter.

10.4 The rank obstruction

The single quantitative difficulty can be named precisely.

Proposition 10.4 (The order-vs-Frobenius gap is exactly $\sqrt{\text{rank}}$). (PROVED) *On a simple Euclidean Jordan algebra of rank r , with order-unit norm $\|a\|_{\text{op}} = \max_i |\lambda_i(a)|$ and Frobenius norm $\|a\|_2 = (\sum_i \lambda_i(a)^2)^{1/2}$,*

$$\|a\|_{\text{op}} \leq \|a\|_2 \leq \sqrt{r} \|a\|_{\text{op}},$$

and the upper ratio $\sqrt{r}$ is attained at $a = \mathbf{1}$. Thus $r = 2$ for spin factors (dimension-free), $r = n$ for $H_n(\mathbb{R}/\mathbb{C}/\mathbb{H})$ (growing), $r = 3$ for the Albert algebra.

The cohomological splitting is known to be dimension-free in the *Frobenius* norm: the Casimir idempotent of the multiplication algebra $M(J)$, written as a Haar average over the compact unitary group $U(M(J))$ of *Frobenius*-isometries, has projective tensor norm ≤ 1 , giving $\|s\|_{\text{F}} = 1/\sigma_{\min}(d^1) = O(1)$ (confirmed numerically: $\dim \text{Ker } d^1 = \dim \text{Der}(J)$ matched exactly, and $1/\sigma_{\min}$ stays ≤ 1 and decreasing for $H_n(\mathbb{R}/\mathbb{C})$ and spin out to large n). This is *not* enough: $U(M(J))$ -isometries are not order-isometries (Proposition 10.3 gives only $\text{Aut}(J)$ for the order norm), so the Frobenius bound need not survive the conversion to the order-unit norm, with a worst-case loss of $\sqrt{r}$. Two caveats forbid cheap fixes:

Remark 10.5 (Why a Frobenius splitting cannot simply be converted). (PROVED; OBSTRUCTION) On $H_n(\mathbb{R})$ the linear map $h(x) = x_{11}\mathbf{1}$ has $\|h\|_{\text{op}} = 1$, $\|h\|_{\text{F}} = \sqrt{n}$, and an order-bounded coboundary $\|d^1 h\|_{\text{op}} \leq 3$, yet every primitive of $d^1 h$ has Frobenius norm $\geq \sqrt{n}$. So an order-bounded coboundary can force a $\sqrt{n}$ Frobenius primitive: one must not seek Frobenius-bounded primitives, only order-bounded ones. Likewise the rank-one decomposition route fails: the identity on the trace-zero part of $H_n(\mathbb{C})$ has nuclear norm $\geq \dim \geq n^2$ while its operator norm is 1, so summing rank-one primitive estimates loses dimension. The splitting must be estimated *directly* in the order norm.

10.5 The exact-adjoint splitting benchmark

The following results (Agent B, 2026-06-05) resolve the obstruction at the level of *exact* cocycles in the *adjoint* module $M = \mathcal{B}$: a dimension-free order-unit-norm splitting of d^1 exists for every simple factor and, by a summand-count-free direct-sum reduction, for every finite-dimensional JB-algebra. This is a major step; it is *not* yet Open Problem 10.1 (see Section 10.6). We mark these (PROVED; EXACT-ADJOINT BENCHMARK).

Proposition 10.6 (Spin factors). (PROVED; EXACT-ADJOINT BENCHMARK) *For a spin factor $V = \mathbb{R}\mathbf{1} \oplus H$ there is an $O(H)$ -equivariant right inverse S of d^1 (adjoint module) with $\|Sf\|_{\text{F}} \leq 2\|f\|_{\text{inj}}$; by the rank-2 norm comparison (Proposition 10.4) the order-unit constant is $\|S\|_{\text{op} \rightarrow \text{op}} \leq 4\sqrt{2}$, independent of $\dim H$. The associated next-arrow estimate also holds: a normalized adjoint 2-cocycle θ with $\theta(\mathbf{1}, z) = 0$ obeys $\text{dist}(\theta, \text{Im } d^1) \leq (2\sqrt{2} + 2)\|J\theta\|$.*

(Independently obtained by both authors; the rank-2 structure makes the order and Frobenius norms $\sqrt{2}$ -equivalent, so the spin gap of Proposition 10.4 is absent.)

Proposition 10.7 (Direct sums). (PROVED; EXACT-ADJOINT BENCHMARK) *If $\mathcal{B} = \bigoplus_r \mathcal{B}_r$ (order-unit norm $= \max_r$) and each factor has an exact adjoint splitting with constant K_r , then $\mathcal{B}$ has one with constant $\max_r K_r + 1$, independent of the number of summands (off-block components are recovered by $P_r f(e_r, \cdot)$, with no sum over r).*

Remark 10.8 (Scope of summand-count independence). (PROVED) Proposition 10.7 is for the adjoint (or block-respecting) module. For an *arbitrary* Jordan module the coboundary need not be block-diagonal even when cross products vanish in $\mathcal{B}$: central idempotents can act with Peirce eigenvalue $\frac{1}{2}$ on a module component (e.g. for $\mathbb{R}e_1 \oplus \mathbb{R}e_2$ the off-diagonal Peirce module of $H_2(\mathbb{R})$ has both e_i acting by $\frac{1}{2}$). Such mixed Peirce- $\frac{1}{2}$ components must be controlled separately; this is the module-side qualification on summand-count independence.

Proposition 10.9 (Commutative scalar modules). (PROVED) *Algebraically, every finite-dimensional unital Jordan $\mathbb{R}^m$ -module splits into one-dimensional modules with action $x \cdot m = \ell(x)m$, where $\ell(x) = x_k$ (coordinate) or $\ell(x) = \frac{1}{2}(x_p + x_q)$ (half-sum). In the associated max-sector norm, the support-unit formula $Sf(x) = f(x, s)$ (with $s = e_k$ resp. $e_p + e_q$) is a norm-one right inverse, and $\Pi = d^1 S$ is a projection onto $\text{Im } d^1$ with $\|\Pi\| \leq 3$. For arbitrary equivalent module norms this statement carries the corresponding complementability constant. In the max-sector norm the next-arrow estimate $\text{dist}(\theta, \text{Im } d^1) \leq 12\|J\theta\|$ holds (sharpened to ≤ 2 for coordinate modules), including the Peirce- $\frac{1}{2}$ half-sum modules.*

The hard case is the high-rank matrix factors, where the rank gap of Proposition 10.4 is real.

Theorem 10.10 (High-rank matrix factors). (BENCHMARK; AGENT B PROOF, RE-AUDIT PENDING) *There is a universal constant C such that for every n and $F \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$: for an exact adjoint coboundary $f = d^1 h$ on $J = H_n(F)$ there is a derivation $\delta \in \text{Der}(J)$ with $\|h - \delta\| \leq C\|f\|_{\text{inj}}$, i.e. d^1 has an order-unit right inverse of norm $\leq C$ uniformly in n . The construction proceeds by: a fixed diagonal-frame Peirce gauge ($\|h|_D\| \leq 11\|f\|$); a coherent off-sector leakage globalization (averaged squared diagonal-sign commutator with spectral gaps $\{0, \frac{1}{2}, 1, \frac{3}{2}\}$); and sector-preserving edge-map reconstruction in each Peirce sector — real-symmetric (constant 60), complex Hermitian (84), complex antilinear (124), and quaternionic — via random-triple matching-curvature averaging. The fixed-frame splitting is frame-covariant (constant ≤ 11 , projection ≤ 33 for every Jordan frame).*

Remark 10.11 (Verification status of Theorem 10.10). (AUDIT CAVEAT) The matrix construction is recent and intricate. Two sidecar audits examined *earlier* formulations of the sector-reconstruction step (a Haagerup/base-vertex route) and found a genuine ordinary-vs-completely-bounded gap there; Agent B's final proof instead uses random-triple matching reconstruction, which is designed to avoid that gap, but the audits did not re-verify the final argument. A logarithmic Schur-multiplier family is the sharpest stress test; its coboundary appears to grow logarithmically as well, so it is not a counterexample. We therefore record Theorem 10.10 as an Agent B proof claim at the exact-adjoint-benchmark level, with independent re-audit of the matching-reconstruction steps still outstanding.

Corollary 10.12 (Master adjoint-benchmark corollary). (BENCHMARK; MODULO REMARK 10.11) *There is a universal constant C such that for every finite-dimensional JB-algebra $\mathcal{B}$ and every exact adjoint coboundary $f = d^1 h \in C^2(\mathcal{B}, \mathcal{B})$ there is a derivation δ with $\|h - \delta\| \leq C\|f\|_{\text{inj}}$: the exact-adjoint coboundary inversion is dimension-free, uniformly over all $\mathcal{B}$ and all numbers of simple summands.*

10.6 The next-arrow estimate and what remains open

The exact-adjoint splitting (Corollary 10.12) is the right inverse for *exact* cocycles. Error reduction additionally needs the *next-arrow* estimate $\text{dist}(\theta, \text{Im } d^1) \leq K'\|J\theta\|$ for *approximate* cocycles, and several further robustness properties. Current status of the next-arrow estimate: proved for commutative scalar modules (Proposition 10.9), for the normalized spin sector (Proposition 10.6), and

for the diagonal-source ($D \times D$) block of matrix factors; *open* for the mixed ($D \times E$) and off-diagonal ($E \times E$) matrix blocks.

Proposition 10.13 (Fixed diagonal-frame matrix next-arrow). (PROVED; FIXED-FRAME BENCHMARK) *Let $D = \mathbb{R}^n$ act on $M = H_n(F)$, $F \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, by the diagonal-frame Jordan action $x \cdot m = x \circ m$. For every unit-normalized symmetric 2-cochain $\theta: D \times D \rightarrow M$,*

$$\text{dist}(\theta, \text{Im } d^1) \leq C \|J\theta\|_{\text{inj}}, \quad \|\theta - \Pi_n \theta\|_{\text{inj}} \leq C' \|J\theta\|_{\text{inj}},$$

where $\Pi_n = d^1 S_n$ is the bounded Rademacher projection onto coboundaries. The proof decomposes the off-diagonal entries into endpoint, one-tail, and tail-tail Walsh pieces and reconstructs the tail-tail component by a two-density sparse-sign identity. This closes the fixed-frame $D \times D$ source block only; it is not the full matrix next-arrow theorem.

For the full high-rank matrix next-arrow problem, a fixed frame splits the source space as $J = D \oplus E$ (diagonal plus off-diagonal Peirce sectors). The $D \times D$ block is Proposition 10.13. The open blocks are $D \times E$, expected to require cochain-level leakage control using diagonal sign spectral gaps, and $E \times E$, expected to require Peirce curvature/matching tests.

Open Problem 10.14 (What separates the benchmark from Open Problem 10.1). (OPEN) *To upgrade Corollary 10.12 to the structure theorem one needs, all with dimension-free order-unit constants: (i) the pre-cohomological construction of approximate idempotent elements, Jordan frames, Peirce splittings, and coordinatization maps from an arbitrary ε -JB algebra; (ii) the next-arrow estimate for arbitrary relevant Jordan modules, not just the adjoint module (the mixed Peirce- $\frac{1}{2}$ components of Remark 10.8 and the matrix $D \times E$, $E \times E$ blocks); (iii) tolerance of approximate module actions, since during the Newton iteration the codomain is itself only an ε -JB algebra; (iv) a stable gauge across iterations; (v) very possibly an incremental assembly (applying error reduction only to controlled subalgebras at each merge/extend step), since naive frame-averaging of the fixed-frame matrix splitting is ruled out — the averaged diagonal compression acts on the traceless part by a factor $\sim 1/n$, a rank-sized loss; and (vi) for the factorization application, a positivity-capable or concrete output (Remark 8.7), which an abstract Jordan isomorphism does not supply.*

In short: the dimension-free order-unit splitting of the Jordan coboundary, long the suspected obstruction, is now available at the exact-adjoint benchmark level for every finite-dimensional JB-algebra (Corollary 10.12); the structure theorem Open Problem 10.1 remains open because the pre-cohomological construction, approximate-cocycle, arbitrary-module, approximate-module, and positivity-output gaps recorded in Open Problem 10.14 are still unresolved. This is genuine and substantial progress over the earlier state, in which even the exact splitting was unresolved beyond small dimension.

11 The exponent: $\sqrt{\eta}$ versus η

The bridge (Section 6) gives an ε -JB algebra with $\varepsilon = O(\sqrt{\eta})$, unconditionally. Kitaev's completely positive theorem achieves the stronger $\varepsilon = O(\eta)$. This section explains the gap, shows that the natural class on which $O(\eta)$ might be recovered is the *decomposable* maps, and records precisely what is proved (a conditional $O(\eta)$ theorem under a dilation-compatibility hypothesis) and what is conjectural (the general decomposable $O(\eta)$ statement), together with the obstructions that make the naive arguments fail.

11.1 One hole versus two holes

Remark 11.1 (Why CP gives η and positivity gives only $\sqrt{\eta}$). (PROVED FOR THE BRIDGE AT $\sqrt{\eta}$; η OPEN IN GENERAL) Both regimes are governed by a single “hole”: the failure of one inserted Φ (or P) to be invisible. For a unital positive Φ a single insertion is controlled only in the seminorm of the state $\omega = \rho \circ \Phi$, where Jordan–Schwarz gives a defect of *squared* size $O(\eta)$, hence norm $O(\sqrt{\eta})$ — this is the source of the unconditional bridge exponent. Kitaev’s $O(\eta)$ does *not* come from a single insertion: the associativity defect is an intrinsically *bilinear* pairing of *two* holes living in the Stinespring dilation. In his notation the one-hole vector $R_k(X)$ satisfies $R_k(X)^\dagger R_k(X) = \Phi^2(\Phi(X^\dagger)\Phi(X)) - \Phi(\Phi^2(X^\dagger)\Phi^2(X)) = O(\eta)\|X\|^2$, so $\|R_k(X)\| = O(\sqrt{\eta})\|X\|$, while the defect equals a product $W = V_1^\dagger R_1(X^\dagger)^\dagger u_{3,0}(\Phi(Y)) V_3 R_0(Z)$ of two such vectors, whence $\|W\| = O(\sqrt{\eta}) \cdot O(\sqrt{\eta}) = O(\eta)$ by Cauchy–Schwarz. The two-hole pairing requires complete positivity — it lives in the dilation Hilbert space and uses matrix amplifications $\text{id}_n \otimes \Phi$ — which a general positive map lacks. Only the single-insertion $O(\sqrt{\eta})$ bound survives unconditionally.

A two-hole structure reappears precisely when each summand of Φ carries a dilation, i.e. for decomposable maps.

11.2 Decomposable maps and the norm correction

Definition 11.2 (Decomposable map). (CITED) Let τ denote the transpose (a Jordan automorphism of $B(\mathcal{H})_{\text{sa}}$). A linear map $\Phi: B(\mathcal{H}) \rightarrow B(\mathcal{H})$ is *decomposable* if it admits a decomposition

$$\Phi = \Phi_0 + \Psi_0 \circ \tau$$

with Φ_0, Ψ_0 completely positive (CP); equivalently Φ is a sum of a CP map and a co-CP map.

Proposition 11.3 (The standard decomposable norm is the wrong hypothesis). (PROVED) *The Haagerup–Wittstock decomposable norm is not 1 for every unital positive decomposable map: the transpose τ_n on M_n is unital, positive and decomposable, but $\|\tau_n\|_{\text{dec}} = \|\tau_n\|_{\text{cb}} = n$ (apply $\text{id}_n \otimes \tau_n$ to the flip). The correct boundedness input is instead the component bound: for a unital map with a fixed decomposition $\Phi = \Phi_0 + \Psi_0 \circ \tau$, unitality gives $\Phi_0(\mathbf{1}) + \Psi_0(\mathbf{1}) = \mathbf{1}$, so $\|\Phi_0\|_{\text{cb}}, \|\Psi_0\|_{\text{cb}} \leq 1$. This per-summand budget is the right hypothesis; the total budget 2 is not the same as a single dilation.*

11.3 A conditional $O(\eta)$ theorem, and why it is conditional

There is a valid Kitaev-strength theorem under an explicit *dilation-compatibility* hypothesis that supplies the missing two-hole structure.

Theorem 11.4 (Dilation-compatible $O(\eta)$ bridge). (PROVED, CONDITIONAL ON THE LIFTED-UCP HYPOTHESIS) *✓ af-validated Suppose there are a finite-dimensional Hilbert space K , a unital order-isometric Jordan embedding $j: B(\mathcal{H})_{\text{sa}} \rightarrow B(K)_{\text{sa}}$, and a unital positive map $C: B(K)_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ with $\Phi = Cj$, such that the lift $F = jC$ extends to a unital completely positive map $B(K) \rightarrow B(K)$ with $\|F^2 - F\|_{\text{cb}} \leq \eta$. Then, with $P = \theta(2\Phi - \mathbf{1})$, $A = \text{Im } P$, $a \bullet b = P(a \circ b)$, the algebra A is an $O(\eta)$ - ε -JB order-unit algebra.*

Proof idea. F is UCP and η -idempotent, so Kitaev’s completely positive theorem applies to F and makes its near-fixed range an $O(\eta)$ approximate C^* -algebra; the symmetrised product is $O(\eta)$ - ε -JB. The intertwining $Fj = j\Phi$ (hence $Fj = jP$ by functional calculus) embeds A isometrically into that structure, and the ε -JB estimates restrict along the embedding. Each of the three steps is independently machine-validated: idempotence inheritance $\|\Phi^2 - \Phi\| \leq \|F^2 - F\|$ (af), the spectral

intertwining $\theta(2F - \mathbf{1})j = j\theta(2\Phi - \mathbf{1})$ (af), and the $C^* \rightarrow \text{JB}$ symmetrisation that supplies the $O(\eta)$ crux (af). $\square$

The hypothesis $\|F^2 - F\|_{\text{cb}} \leq \eta$ is *not* automatic from a CP+coCP decomposition. Two obstructions show this.

Proposition 11.5 (Doubling obstruction). (PROVED; OBSTRUCTION) *For $\Phi = Cj$ one computes $F^2 - F = j((\Phi - \mathbf{1})C)$, and $\|\Phi^2 - \Phi\|$ controls $(\Phi - \mathbf{1})$ only on inputs $\Phi(x)$, not on all of $C(D)$. Concretely, with $j: \ell_2^\infty \rightarrow \ell_4^\infty$, $j(x_1, x_2) = (x_1, x_2, x_1, x_2)$ and $C = \begin{pmatrix} \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$, the map $\Phi = Cj = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ is exactly idempotent, yet $F = jC$ has $\|F^2 - F\|_{\infty \rightarrow \infty} = 1$. So exact idempotence of Φ does not imply approximate idempotence of the lift.*

Proposition 11.6 (Bad decompositions of an exact projection). (PROVED; OBSTRUCTION) *For the depolarizing projection $D(x) = \frac{1}{2} \text{Tr}(x)\mathbf{1}$ on M_2 (so $\eta = 0$) and the decomposition $\Phi_0 = \varepsilon R$, $\Psi_0 = D - \varepsilon R$ with $R(x) = \langle 0|x|0 \rangle |0\rangle\langle 0|$, both summands are CP and $\Phi = \Phi_0 + \Psi_0 \circ \tau = D$, yet the natural lifted map has off-diagonal universal-envelope defect $\geq \varepsilon/2$ for every $\varepsilon > 0$. Hence no estimate controlling the lift can follow from an arbitrary supplied CP+coCP decomposition.*

(Both obstructions concern the *proof strategy*, not the conclusion: for these small examples the actual Jordan defect is zero, since any unital idempotent on M_2 yields an exact spin-factor product. A genuine failure of $O(\eta)$ would need matrix size ≥ 3 , where no counterexample is known.)

11.4 The general decomposable conjecture

Open Problem 11.7 (Decomposable $O(\eta)$). (OPEN; CONJECTURAL) *If Φ is unital positive and almost idempotent with a controlled CP+coCP decomposition $\Phi = \Phi_0 + \Psi_0 \circ \tau$ ($\Phi_0(\mathbf{1}) + \Psi_0(\mathbf{1}) = \mathbf{1}$), then the bridge exponent improves from $O(\sqrt{\eta})$ to $O(\eta)$.*

By Propositions 11.3, 11.5 and 11.6, a proof must either select a compatible dilation/lift satisfying the lifted almost-idempotence estimate of Theorem 11.4, or establish a direct intrinsic CP/co-CP *two-hole cancellation* for the Jordan defect using only $\|\Phi^2 - \Phi\|$ — expressing the defect as a Cauchy–Schwarz pairing of two $O(\sqrt{\eta})$ holes, as in Remark 11.1. Neither is currently in hand. The exponent thus tracks the presence of a dilation: complete positivity (= a dilation) gives η ; mere positivity gives $\sqrt{\eta}$; decomposability gives η exactly when its CP/co-CP dilations can be made compatible.

12 Status, relation to van Luijk–Wilming, and open problems

This section records the overall architecture, places it against the exact theory it approximates, and lists what is proved, cited, and open. The development has five strands: the *bridge* (Section 6), turning an almost-idempotent positive map into an approximate Jordan algebra; the *exact factorization* (Section 8) and its *commutative* core (Section 9); the *abstract structure theorem* (Section 10); and the *exponent* question (Section 11). Throughout, $\Phi: B(\mathcal{H})_{\text{sa}} \rightarrow B(\mathcal{H})_{\text{sa}}$ is unital positive with $\|\Phi^2 - \Phi\| \leq \eta$, $P = \theta(2\Phi - \mathbf{1})$, $A = \text{Im } P$, and $a \bullet b = P(a \circ b)$.

12.1 Relation to van Luijk–Wilming

The structures here are the approximate counterparts of an exact object from the theory of quantum statistical experiments.

Remark 12.1 (van Luijk–Wilming minimal sufficient Jordan algebra). (CITED) For a faithful statistical experiment (ρ_θ) , the *minimal sufficient J^* -algebra* is

$$J_{(\rho_\theta)} = \bigcap \{ \text{Fix}(T) : T \text{ a } (\rho_\theta)\text{-preserving unital positive map on } L(\mathcal{H}) \},$$

an exact intersection of fixed-point sets of unital positive maps, and it “is generated by the Neyman–Pearson tests” $[\rho > t\sigma]$ [8]. Our near-fixed-point algebra $A = \text{Im } P$ is the *approximate* analogue: P replaces an exact conditional expectation onto a fixed-point set by an approximate idempotent, and $a \bullet b = P(a \circ b)$ replaces the exact Effros–Størmer product $r \bullet s = P(r \circ s)$ of Theorem 8.1. In their setting the Jordan algebra is always special (concretely realised; exceptional $H_3(\mathbb{O})$ excluded); the bridge target inherits this. The faithful-state hypothesis in their fixed-point theory is qualitative; as Section 7 shows, its approximate analogue is governed by a quantitative conditioning of the invariant state and does not, by mere existence, transfer.

12.2 What is proved, and the three genuinely open problems

The headline results of this report are (i) the unconditional bridge (Theorem 6.1), turning a unital positive almost-idempotent map into an $O(\sqrt{\eta})$ -JB algebra with no dilation or complete positivity; and (ii) the reduction of the entire remaining factorization problem to a single, sharply-stated stability question (Open Problem 8.5), via the conditional exact factorization Theorem 8.6. Beyond these, the abstract structure theory has advanced to a dimension-free *exact-adjoint splitting benchmark* (Corollary 10.12).

The open problems are independent and of three different flavours.

- **The exponent** (Section 11): whether decomposability restores the two-hole mechanism and upgrades $\sqrt{\eta}$ to η . Proved under a dilation-compatibility hypothesis (Theorem 11.4); conjectural in general (Open Problem 11.7); the naive doubled proof is obstructed (Propositions 11.5 and 11.6).
- **Exact factorization** (Sections 8 and 9): whether a near-positive idempotent is $O(\sqrt{\eta})$ -close to a genuine positive idempotent (Open Problem 8.5). Generic positivity rounding is ruled out (Proposition 8.4); the commutative case reduces to a stochastic stability conjecture (Open Problem 9.2) with many proved special cases (Section 9) and a single remaining geometric lemma (Open Problem 9.15).
- **The abstract structure theorem** (Section 10): whether every ε -JB algebra is $O(\varepsilon)$ -close to a genuine JB-algebra, dimension-free (Open Problem 10.1). The exact-adjoint splitting is now available as a benchmark (Corollary 10.12); the gap includes the pre-cohomological construction of frames, Peirce splittings, and coordinates, plus approximate-cocycle and arbitrary-module estimates, approximate-module robustness, and positivity-capable output (Open Problem 10.14).

12.3 Status ledger

Table 2 summarises every component. Status tags: (PROVED) = complete proof in this report or transcribed verified proof; (PROVED, COND.) = implication proved, hypothesis open; (BENCHMARK) = proved at the exact-adjoint-cocycle level only; (CITED) = from the literature; (OPEN) = target; (OBSTRUCTION) = proved no-go for a route; (SOURCE-EXTRACTED) = local extraction awaiting primary byte-check; (NUMERICAL) = evidence not used as proof.

A green ✓ in the final column marks results that are additionally *machine-validated* in the adversarial proof framework (af): each such result has a tiny formal workspace whose root statement

is byte-identical to its contract and whose every step is checked by a prover and an independent verifier. In the body these carry a green ‘✓ af-validated’ badge linking to the corresponding workspace `proofs/<id>/`. Thirteen results are af-validated: the bridge theorem and its nine constituent lemmas (Section 6, Section 4) and the dilation-compatible theorem with its three sub-lemmas (Section 11). The full machine-checked dependency graph is Figure 1.

12.4 Outlook

The picture that emerges is uniform. Mere positivity supplies Kadison’s inequality and hence a Jordan, not associative, structure; it supplies one hole of control, hence the exponent $\sqrt{\eta}$; and it withholds the dilation that would supply the second hole, the genuine positivity of P , and the cb-norm-driven dimension-free homotopy. Each open problem is exactly the cost of one missing piece of complete positivity: the exponent (Open Problem 11.7) is the missing second hole; exact factorization (Open Problem 8.5) is the missing positivity of P ; the abstract structure constant (Open Problem 10.14) is the missing cb-norm/order-unit homotopy, now sharpened by the exact-adjoint benchmark but still requiring several pre-cohomological and approximate-module components. None of the three is known to fail. Each has been reduced to sharply stated remaining packages, not to an unconditional theorem.

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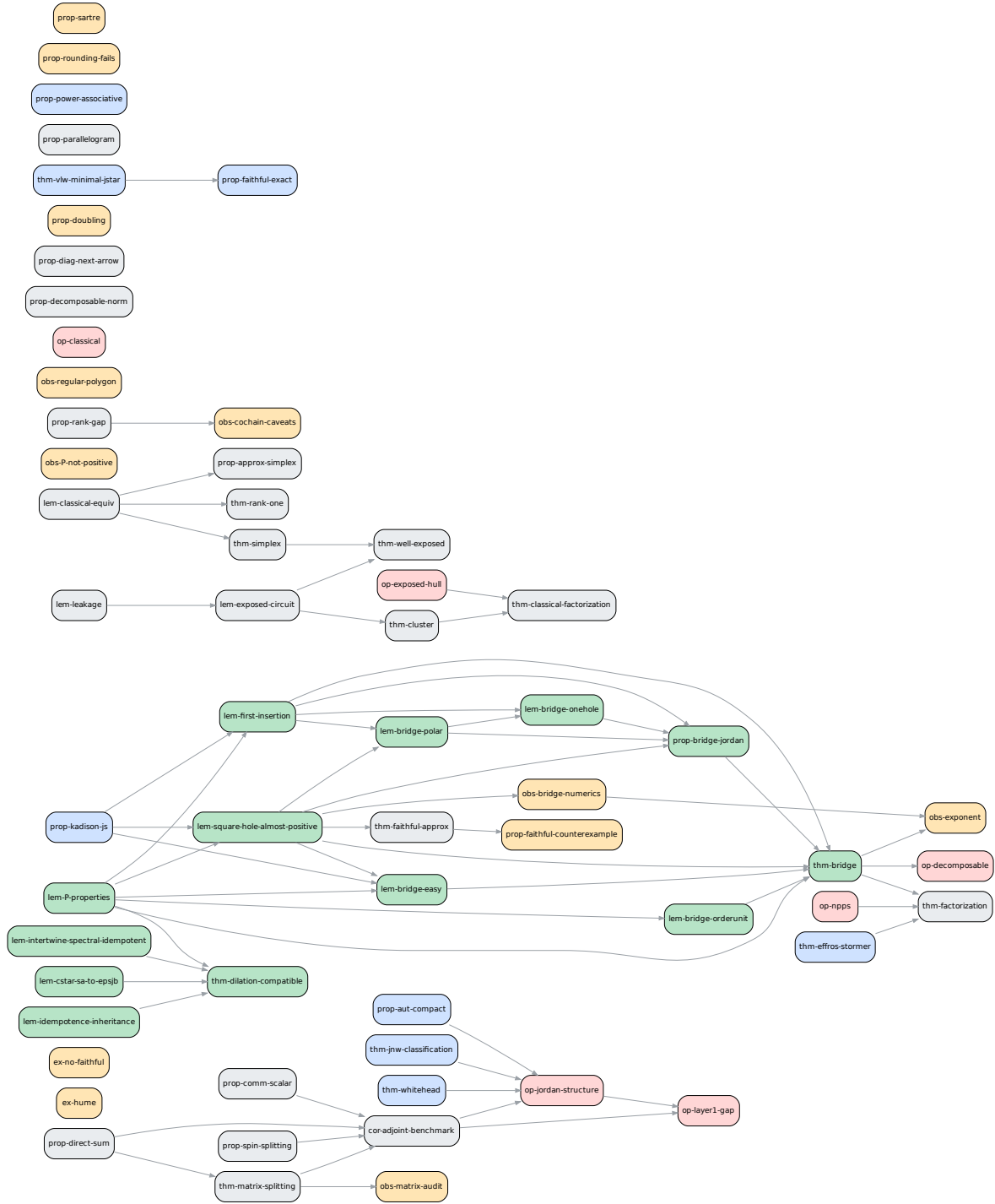


Figure 1: The argument dependency graph, generated from the registry `argument/lemmas/` by `scripts/gen-dag-figure.py`: one node per result, each edge a dependency. Node colour: ■ green = af-validated (machine-checked); grey = proved in prose but not yet formalised; blue = cited from the literature; red = open problem; amber = obstruction / no-go. The PDF is vector graphics: zoom in a reader to inspect individual node labels.

Component	Status	Where	af
Bridge: approx. Jordan algebra at $\sqrt{\eta}$, dimension-free, CP-free	proved	Theorem 6.1	✓
Ambient product under a faithful invariant state ($O(\eta/\lambda)$; bare existence insufficient)	proved	Section 7	
Effros–Størmer exact template	cited	Theorem 8.1	
P need not be positive	proved	Remark 8.2	
Generic positivity rounding fails (spin-factor)	obstruction	Proposition 8.4	
Exact UP factorization (Theorem C)	proved, cond. (Open Problem 8.5)	Theorem 8.6	
Near-positive projection stability	open	Open Problem 8.5	
Classical reduction; Hume sharpness	proved	Lemma 9.3, Example 9.4	
Classical special cases (rank-one, line-segment, simplex, well-exposed, cluster)	proved	Theorem 9.8–9.11	
Exact commutative factorization (cluster geometry)	proved, cond. geom.	Theorem 9.12	
Classical projection stability (general); global exposed-hull	open	Open Problem 9.2, Open Problem 9.15	
Order-vs-Frobenius gap = $\sqrt{\text{rank}}$	proved	Proposition 10.4	
Exact-adjoint splitting: spin, direct sums, commutative scalar	benchmark	Proposition 10.6–10.9	
Exact-adjoint splitting: high-rank matrix factors	benchmark (re-audit pending)	Theorem 10.10	
Master exact-adjoint coboundary inversion	benchmark	Corollary 10.12	
Abstract structure theorem (Layer 1)	open	Open Problem 10.1, Open Problem 10.14	
Exponent η (dilation-compatible)	proved, cond.	Theorem 11.4	✓
Exponent η (general decomposable)	open	Open Problem 11.7	
Whitehead lemmas; $\text{Aut}(J)$ compact	cited (extract.)	Theorem 10.2, Proposition 10.3	

Table 2: Status of the components of the programme. A green ✓ marks a component that is machine-validated in the af proof framework (Figure 1).